Highlights

zgReal-Time Redesign of Public Transport Lines under Fluctuating Traffic ConditionsPredict-then-Optimize Framework for Public Transport Line Redesign under Fluctuating Traffic Conditions

- Existing bus lines are redesigned an hour ahead, only upon large forecast deviations.
- An explicit continuity objective limits disruption for passengers and operators.
- A predict-then-optimize loop: optimize on noisy forecasts, score on realized times.

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ABSTRACT

Public Transport (PT) lines are traditionally designed to optimize performance under nominal traffic conditions. In practice, however, operating conditions frequently deviate from the nominal ones, leading to substantial performance deterioration. Existing adaptation mechanisms typically rely on corrective and reactive interventions, such as stop-skipping, which prove insufficient under large or recurrent traffic fluctuations. In such contexts, incremental adjustments may be insufficient. This paper evaluates the potential of deeper structural redesigns to preserve performance. To this aim, this paper proposes a method to proactively redesign appropriate parts of PT networks when faced with high traffic fluctuations that would otherwise deteriorate operator- and user- performance. We adopt a predict-then-optimize paradigm in which PT lines are reconfigured based on traffic forecasts via an algorithm based on the Non-dominated Sorting Genetic Algorithm III (NSGA-III). For operational feasibility, to avoid excessive structural changes, we enforce high Jaccard edge overlap between the original and redesigned networks. To account for the effect of prediction inaccuracies on redesign quality, we construct a statistical model of the errors made by a well-established deep learning-based prediction model (Diffusion Convolutional Recurrent Neural Network) trained on real-world data. zgComputational results on Mandl's benchmark network and on the large-scale Beijing network show that the controlled redesign of PT lines can bring significant user-centric performance improvement and operational cost reductions in high traffic fluctuation scenarios, while limiting topological modifications.Computational results on Mandl's benchmark network and on the large-scale Beijing network show that the controlled redesign of PT lines brings significant user-centric performance improvement and operational cost reductions in high traffic fluctuation scenarios, while limiting topological modifications. On the Beijing network under high variability, average travel time improves by up to 25.8% while preserving over 85% route overlap; unlike stop-skipping baselines, which break connectivity for a large share of OD pairs, our redesign preserves full OD connectivity. This indicates the potential of transitioning from current static planning paradigms toward continuous and adaptive PT network design.

1. Introduction

Public Transport (PT) networks are typically planned for nominal, scheduled conditions (Ibarra-Rojas, Delgado, Giesen and Muñoz, 2015), while real-world operations are continuously subject to variability caused by congestion and other factors (Büchel and Corman, 2020a; Danaher, Wensley, Dunham, Orosz, Avery, Cobb, Watkins, Queen, Berrebi, Connor et al., 2020), especially in mixed traffic environments. Such routine perturbations occur daily and can propagate across the network, progressively degrading service reliability (Marra and Corman, 2020). For example, in Xi'an, 74.34% of bus runs were delayed (Zhang, Xu, Lu and Fan, 2022). Beyond operational impacts, insufficiently reliable public transport can also induce behavioral responses, including increased reliance on private vehicles (Rodier, Zhang, Harold and Drake, 2025).

Existing approaches to cope with recurrent perturbations can be broadly classified as preventive, reactive, or proactive, depending on the decision horizon and intervention level. Preventive strategies enhance robustness ex ante through structural measures, such as dedicated bus lanes that isolate buses from mixed traffic (Russo et al., 2022). However, limited street space and high infrastructure costs often make such measures infeasible in dense urban areas (Li, Song, Li, Zhang and Miller, 2009), and preventive capacity tends to be inflexible and underutilized (International Transport Forum (ITF), 2024, 2025). Reactive strategies attempt, at the operational level, to restore service after perturbations have occurred, for instance via stop-skipping, holding, or detouring (Peled, Lee, Jiang, Dauwels and Pereira, 2021; Tang, Ceder and Ge, 2018; Zheng, Sun, Kang, Dai and Wu, 2023). These actions deteriorate under

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high uncertainty, inconvenience passengers (Gkiotsalitis and Cats, 2021, §2), and are insufficient to manage cascading perturbations when no alternative network structure is available (Mahdavi and Bhouri, 2024).

A key limitation shared by both preventive and reactive strategies is that the PT network structure typically remains fixed, despite persistent changes in operating conditions. Network design decisions are revised infrequently, and traditional models assume deterministic or fixed travel times (Fonseca, 2020; Büchel and Corman, 2020b), limiting their ability to address recurrent traffic fluctuations. Proactive approaches aim to overcome this limitation by anticipating variability through short-term predictions and revising the line design before the next operating interval (Peled, Lee, Jiang, Dauwels and Pereira, 2019).

The **main contribution** of this paper is to assess the potential of predict-then-optimize proactive redesign of bus networks. To this aim, we propose a new method that adapts bus lines to time-varying stochastic traffic congestion. We solve a multi-objective problem to minimize operational cost and user travel time, and to limit structural network modifications, so as to ease operations and user routing. Adaptation is triggered only when forecasted conditions severely deviate from the nominal ones; moreover, such adaptation is anticipated (e.g., 1h in advance) to avoid disrupting journeys of current travelers.

zgResults on the Mandl benchmark and Beijing networks show that notable improvements in user- and operator-centric performance can be achieved while limiting the extent of structural modifications. In the Beijing large scale network, beyond a one-off static re-optimization gain of 11.4%, with no link travel time variability, the benefits grow with such a variability. When it is very high, average travel time improves by up to 25.8% and operational cost by up to 22.6% at with high traffic variability, while route overlap stays above 85% (Table 5). Compared with a stop-skipping baseline, which improves travel time by at most 19.7% but only by disconnecting 28–39% of OD pairs under high variability, our redesign attains a larger gain while keeping every OD pair reachable. These findings highlight the potential of proactive bus-line redesign in future public transport systems, where adaptive operational strategies may become easier to deploy thanks to vehicle automation (Fagnant and Kockelman, 2015) and to users' growing familiarity with flexible, app-mediated mobility services, as illustrated by the diffusion of on-demand platforms.

The remainder of the paper is organized as follows. Section 2 reviews related work, Section 3 formulates the problem, Section 4 presents the solution approach, Section 5 describes the evaluation method, Section 6 reports the results, and Section 7 concludes.

2. Related Work

Previous work related to ours typically focuses on either *network design* at strategic and tactical levels, where PT lines are planned from scratch, or *reactive operational control*, which manages daily operations without altering the network structure. *Network redesign* occupies an intermediate position by adapting existing networks rather than starting anew. Within this domain, *offline redesign* relies on re-optimization for known future scenarios, whereas emerging *proactive redesign* approaches anticipate short-term uncertainty. However, these proactive methods often overlook the critical trade-off between responsiveness and structural stability, a gap that this paper specifically addresses.

2.1. Public Transport Network Design under Nominal Conditions

The PT Network Design Problem (TNDP) consists of designing line routes and their frequencies, balancing user- and operator-centric objectives, and has been studied since the foundational formulations of Ceder and Wilson (1986) through successive reviews (Guihaire and Hao, 2008; Kepaptsoglou and Karlaftis, 2009; Durán-Micco and Vansteenwegen, 2022) and reference texts (Desaulniers and Hickman, 2007; Ceder, 2016),Gkiotsalitis (2020).

Most of these approaches formulate public transport networks from scratch, without reference to any pre-existing service configuration; although some formulations begin from a predefined set of candidate lines (Cervantes-Sanmiguel, Chavez-Hernandez and Ibarra-Rojas, 2023), the network topology is typically treated as a fresh design decision rather than an adaptation of an existing operational configuration. These methods are primarily intended for long-term strategic or tactical planning. The TNDP is NP-hard (Magnanti and Wong, 1984).

User-centric performance has been expressed in numerous ways. In addition to overall travel duration (John, Mumford and Lewis, 2014; Heyken Soares, Mumford, Amponsah and Mao, 2019), recent studies also examine network coverage (Cadarso, Codina, Escudero and Marín, 2017), route efficiency (Suman and Bolia, 2019), and demand fulfillment (Barahimi, Eydi and Aghaie, 2021). Operator-centric objectives are focused on limiting operational and capital expenditures (Ahern, Paz and Corry, 2022; Chen, Liu, Wang and Chen, 2018; Calabrò, Araldo, Oh, Seshadri,

Inturri and Ben-Akiva, 2023), or proxies for them, such as the overall length of line routes. Balancing these objectives draws on two distinct methodological strands. Multi-level formulations capture hierarchical decision-making among different agents: Szeto and Jiang (2014) established a bilevel framework in which an upper-level planner minimizes passenger transfers and a lower-level model handles PT assignment; Tian, Wang and Lin (2021) extended this to a trilevel structure with passenger routing strategies at an intermediate level. Multi-objective formulations (as ours) instead seek Pareto-optimal trade-off solutions across conflicting criteria: Cervantes-Sanmiguel et al. (2023) analyzed the trade-off between travel-time reduction and fare reduction.

Note that all these methods design PT network based on nominal (e.g., average, peak, etc.) traffic conditions. Therefore, as noted by (Tan, Peng, Huang, Wang, Qiu and Liu, 2024), the resulting networks may be inefficient faced with real-time variations in traffic patterns, deviating from nominal conditions.

2.2. Reactive Operational Control under Disruptions and Uncertainty

In most real-world operations, the structure of the public transport network remains fixed as originally planned. When deviations from nominal conditions occur due to exogenous factors (Nissen, Becker, Dähne, Rabe, Scheffler, Solle and Ulbrich, 2020; Chen, Yu, Zhang and Guo, 2009), operators rely on short-term corrective actions to mitigate supply–demand mismatches. Typical examples include stop-skipping (Peled et al., 2021), short-turning, speed changing, holding (Tang et al., 2018; Manasra and Toledo, 2019; Gkiotsalitis, 2020; Olvera-Toscano, Ríos-Solís, Ríos-Mercado and Sánchez Nigenda, 2023), detouring of interrupted lines (Zheng et al., 2023), and replacement buses. Each control action has distinct disadvantages. Namely, bus holding increases the inconvenience of on-board passengers at the holding stops, short-turning requires passengers to disembark and wait for a subsequent bus trip, rescheduling affects the timetables and the crew/vehicle schedules; speed control cannot be implemented when the required speeds are beyond the safety limits, and stop-skipping results in refused passenger boardings (Gkiotsalitis, 2021). While some works integrate real-time data into operational decision-making, such as schedule adjustment for enhanced reliability (Cao, Ceder and Zhang, 2019), these frameworks focus on timetable or speed control within a fixed route structure and do not explicitly consider structural reconfiguration of the network topology under time-varying conditions. It should be noted that proactive network redesign, as proposed in this work, may generate analogous and in some respects stronger inconveniences on the operational side, affecting vehicle scheduling and driver rostering, in particular with human-driven PT vehicles; it is thus important to limit excessive structural instability (as introduced in §3.4.1).

More importantly, when actual conditions deviate significantly from the nominal ones, such corrective actions may not be sufficient to restore performance (Guo, Araldo and Yacoubi, 2024), since the overall structure is maintained as planned. Strategies such as stop-skipping and short-turning have also appeared in the context of offline or preplanned redesign, to generate alternative route configurations or long-term service plans under hypothetical disruption scenarios. They are outside the scope of this paper, which instead focuses on proactive redesigning lines for the hour ahead based on short-term forecasts.

2.3. Offline and Preplanned Network Redesign under Known Scenarios

Various approaches can be employed in the bus route adjustment problem to generate adaptive bus routes, including rerouting, short-turning, skip-stopping, and branching (Cao and Ceder, 2019; Ma, Yang and Li, 2023; Wu, Kulcsár, Qu et al., 2021). In contrast to the network design approaches described in §2.1, these studies explicitly start from an existing public transport network and allow service or structural modifications under a limited set of anticipated scenarios. Previous research on bus route localized adjustments has often aimed at two main objectives: accommodating demand fluctuations and enhancing integration with subway networks.

Guan, Xiang, Wang, Yan and Zhao (2023) introduced a bilevel planning model to optimize customized bus routes offline based on a spatiotemporal state network. While effective for static planning, this method is impractical for real-time dispatching. Concretely, it relies on a computationally expensive nested algorithm and requires a full re-solve for any new event. Zhang, Jia, Bai, Mao, Ma and Zhang (2018b) investigated long-distance replacement bus services to improve coordination with urban rail networks. While effective under static demand assumptions, this approach is inherently offline and cannot be applied in real-time operational contexts. When passenger flow control is implemented in a subway system, Zhou, Hu, Huang and Deng (2021) optimized existing bus routes to transport passengers temporarily restricted from entry, in case of disruptions. Wang, Chen, Yu, An and Wu (2022) proposed a two-stage method to mitigate disruptions to bus routes under planned construction scenarios, relying on predefined

work-zone impacts and static travel-time estimates. Wu, Zuo, Zhou, Wan, Zhao and Yang (2024) proposed a multi-objective ant colony algorithm to adjust existing bus routes and introduced a sequence-level similarity to prevent the redesigned network from being excessively different from the original. However, their approach remains static and cannot capture time-varying adaptations. Zheng, Sun, Wu and Kang (2024) proposed localized adaptive bus routes under anticipated disruptions, while Zhang, Xu and Wang (2024) addressed large-scale emergency disruptions assuming available spare capacity. These assumptions do not hold under recurrent traffic fluctuations, where no redundant lines exist and adaptation must occur within the same constrained topology.

2.4. Network Redesign under Predictive Uncertainty and Structural Consistency

Traditional network design handles uncertainty primarily by optimizing for expected scenarios (stochastic optimization) or worst-case bounds (robust optimization) (Bertsimas, Gupta and Kallus, 2018; Wang and Qi, 2020); see Ge, Voß and Xie (2022) for a recent review of robustness and disturbances in public transport. However, these approaches typically determine a single network configuration before operations begin, with limited adaptation to realized road conditions (i.e., without reconfiguration). While this ensures stability and ease of operations, it may become ineffective in preserving performance faced with high traffic fluctuations. Another line of work studies topological metrics to evaluate network resilience (Massobrio and Cats, 2024; Pu, Li and Ma, 2022), e.g., they measure vulnerability after a perturbation occurs. These approaches remain purely diagnostic and descriptive, rather than being prescriptive (as the approach we present in this paper), in that they do not provide guidance on the optimization of the network design.

2.5. Vehicle Routing Problems and Demand-Responsive Transport (DRT)

In Vehicle Routing Problems (VRPs) and DRT, vehicle routes are constructed on-demand and solution stability has been employed to keep revised plans close to a baseline (Touzout, Ladier and Hadj-Hamou, 2021). However, this stability applies to routing or scheduling assignments rather than the structure of the PT line network itself. Moreover, DRT and other VRP-based systems do not exhibit a structured network of lines, and therefore fail to achieve sufficient passenger consolidation (Calabrò, Le Pira, Giuffrida, Inturri, Ignaccolo and Correia, 2022). Hybrid fixed/flexible services partly bridge this gap (Calabrò et al. (2023)), but still keep the fixed lines inflexible and flexible routes unstructured. By contrast, we study a high-capacity mass-transit system organized around a structured network of fixed lines, where existing lines are reconfigured rather than routes being computed from scratch. Our approach is therefore conceptually distinct from VRP- or DRT-based formulations. A direct comparison with DRT is also inappropriate, since we address high demand levels typical of dense urban cores, for which DRT is generally unsuitable in terms of both computational scalability and service performance (Calabrò et al., 2022).

2.6. Research gaps

Table 1 summarizes related work. Most existing studies treat PT network design as a deterministic optimization problem, ignoring the inherent uncertainty and variability in urban traffic conditions. Only a few studies consider any form of uncertainty, and few redesign the network in response to short-term forecasts while also modeling travel-time variability in a systematic way. An exception is Guo, Araldo and El Yacoubi (2025), where, however, (i) future travel times on road links were assumed to be perfectly known (which is unrealistic), so the impact of prediction errors was not considered, while we model the error of a well established prediction model, (ii) only a single simplified objective was adopted, while we solve a multiobjective optimization, and (iii) their approach is tested only on a simple hypothetical scenario, while we consider real-world dataset to fit the prediction error models.

To our knowledge, this work is the first to systematically study the potential of proactively redesigning PT lines (e.g., for the hour ahead) under fluctuating traffic. Redesign decisions are based on short-term forecasts, are triggered when needed (i.e., when traffic conditions are highly perturbed causing user- and operator-performance degradation), and balance passenger travel time against operational cost through controlled structural changes.

3. Model and Problem Formulation

As illustrated in Fig. 1, our methodology follows a predict-then-optimize framework that explicitly integrates current observations into prediction. The prediction model is first trained offline on historical data to learn travel times on road links. During operation, it receives current observations from the latest time steps and outputs a short-term prediction of link travel times $\hat{t}_{ij}$, which are then supplied to the optimization model for potential network redesign. To ensure timely responsiveness, this procedure is repeated periodically, i.e., once per hour. At the start of each time

Study	Solution Method	Online/Proactive Redesign	Similarity Control	Decision Focus
(Magnanti and Wong, 1984)	Exact methods	No	No	ND
(John et al., 2014)	Multi-objective GA	No	No	ND
(Heyken Soares et al., 2019)	Scaled network GA	No	No	ND
(Cadarso et al., 2017)	Branch-and-cut	No	No	ND
(Ahern et al., 2022)	Gradient-based	No	No	ND
(Tan et al., 2024)	Congestion analysis	No	No	FS
(Guan et al., 2023)	Custom bilevel	No	No	OR
(Zhang et al., 2018b)	Tailored GA	No	No	OR
(Wang et al., 2022)	Two-stage heuristic	No	No	DR
(Wu et al., 2024)	Ant Colony	No	NW	OR
(Zheng et al., 2024)	Heuristic	No	No	DR
(Cao and Ceder, 2019)	Skip-stop heuristic	No	No	VC
(Ma et al., 2023)	Integer programming	No	No	OR
(Tang et al., 2018)	Optimal control	No	No	VC
(Manasra and Toledo, 2019)	Branch-and-bound	No	No	VC
(Gkiotsalitis, 2020)	Evolutionary	No	No	VC
(Olvera-Toscano et al., 2023)	Holding optimization	No	No	VC
(Zheng et al., 2023)	Coordination heuristic	No	No	VC
(Guo et al., 2025)	Exact methods	Partial	Penalty	NR
Our Approach	**NSGA-III based optimization**	**Yes**	**Jaccard**	**NR**

Notes: Online/Proactive Redesign – **No**: no online or proactive redesign; **Partial**: limited redesign capability (e.g., sampling-based or scenario-specific updates); **Yes**: proactive redesign enabled (e.g., predict-then-optimize). Similarity Control – **No**: no explicit similarity mechanism; **NW**: Needleman–Wunsch-based similarity; **Penalty**: penalty-based similarity; **Jaccard**: Jaccard-based similarity. Decision Focus – **ND**: from-scratch network design; **FS**: frequency setting; **OR**: offline or preplanned redesign of an existing network; **DR**: perturbation-oriented redesign or response; **VC**: vehicle or operational control; **NR**: network redesign under uncertain travel times.

Table 1
Comparative analysis of public transport optimization approaches across decision scope, online/proactive redesign capability, and similarity control mechanisms.

slot, the new network is announced in advance (e.g., 1h), so passengers can plan a route that remains valid for their entire trip and is never changed mid-journey; we assume here that the longest trip is shorter than one time slot (1h). This also keeps the bus routes easier to manage. Further details are given in §3.4.1. The resulting network design is then evaluated against realized travel times t_{ij} to assess the actual performance of our adaptive network under realistic conditions. The notation is summarized in Table 2.

3.1. Graph Model of PT Network

We represent the underlying road infrastructure as an undirected substrate network $\mathcal{G}^{\text{sub}} = (\mathcal{V}^{\text{sub}}, \mathcal{E}^{\text{sub}})$, where $\mathcal{V}^{\text{sub}}$ are candidate bus stops and $\mathcal{E}^{\text{sub}} = \mathcal{V}^{\text{sub}} \times \mathcal{V}^{\text{sub}}$ the potential connections among them. Observe that a single $\ell \in \mathcal{E}^{\text{sub}}$ may represent a sequence of road links from a stop to another. Let $\mathcal{L}$ denote the set of all PT lines in the network. Each PT line $\ell \in \mathcal{L}$ is defined by a set of undirected substrate edges, representing the physical stop-to-stop connections served by the line. To ensure path continuity and avoid disconnected or cyclic line configurations, the undirected edge set of each line is represented through a directed simple path graph $\mathcal{G}_\ell = (\mathcal{V}_\ell, \mathcal{E}_\ell)$ on the undirected substrate $\mathcal{G}^{\text{sub}}$, where $\mathcal{V}_\ell \subseteq \mathcal{V}^{\text{sub}}$ is the set of stops served by line ℓ, and $\mathcal{E}_\ell \subseteq \mathcal{E}^{\text{sub}}$ denotes the set of connections between consecutive stops along the path representation of line ℓ. Vehicles operating on line ℓ are assumed to traverse this path in both directions; the reverse traversal is implicitly captured through symmetric travel times (Schmid, 2014). The complete public transport network is constructed by aggregating all individual lines and is denoted by $\mathcal{G}_{\text{PT}} = (\mathcal{V}_{\text{PT}}, \mathcal{E}_{\text{PT}}, \mathcal{H})$. In this formulation, $\mathcal{V}_{\text{PT}} = \bigcup_{\ell \in \mathcal{L}} \mathcal{V}_\ell$ and $\mathcal{E}_{\text{PT}} = \bigcup_{\ell \in \mathcal{L}} \mathcal{E}_\ell$, while $\mathcal{H} = \{h_\ell \mid \ell \in \mathcal{L}\}$ represents the set of line headways, with each h_ℓ defined as the time interval between two consecutive vehicles (Vuchic, Clarke and Molinero, 1981). Accordingly, $\mathcal{G}_{\text{PT}}$ can be viewed as a line-labeled directed multigraph for modeling purposes, while operationally each line provides bidirectional service along its physical path.

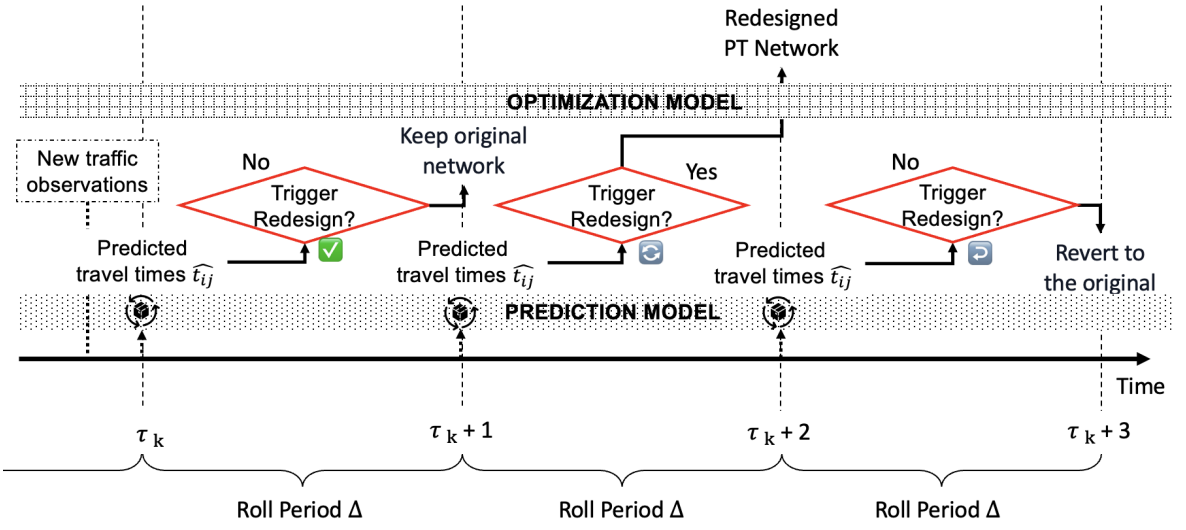


Figure 1: Proactive Line Redesign Framework.

For each PT line $\ell \in \mathcal{L}$, headway h^ℓ can be calculated by $h^\ell = \frac{T^\ell}{n^\ell}$, where T^ℓ represents the round trip time along the entire line, and n^ℓ denotes the number of vehicles operating on that line (Eq. 4.4 of (Vuchic et al., 1981)). Let t^ℓ denote the one-way travel time from the origin terminal to the destination terminal of line ℓ, the round-trip cycle time is $T^\ell = 2t^\ell$. Accordingly, the headway of line ℓ is given by $h^\ell = \frac{2t^\ell}{n^\ell}$. We use a fixed dwell time d, at each stop to account for the time a bus spends at a stop, including the time needed for passengers to board and alight, and the time lost for accelerating and decelerating. While in practice dwell times vary with boarding and alighting counts, constant dwell time is standard at the network design level (Huynh and Barthelemy, 2022; Chen, Li and Qu, 2022; Schöbel and Urban, 2022), where demand is represented as an OD matrix rather than individual trip records and the optimization horizon (one hour) far exceeds individual dwell events. Variable dwell time is primarily relevant in operational-control models (bus bunching, holding) that operate at a finer temporal resolution. Assuming random passenger arrivals, the average waiting time for line ℓ is given by

$$w^\ell = \frac{h^\ell}{2}, \text{ where } h^\ell \text{ denotes the service headway of line } \ell. \tag{1}$$

This assumption is standard in public transport modeling (Parbo, Nielsen and Prato, 2018; Zhang, Wang and Meng, 2018a). Liu and Miller (2020) confirm that such an assumption is appropriate when the headway is relatively small (see their § 3.3.1), and passengers simply go at the stop as soon as possible, without explicitly synchronizing at specific PT departure times; they also show that such a simple behavior is most time the most convenient for users, even better than a more complicated possible user strategies. For a transfer at stop i from line ℓ to line ℓ', the generalized transfer cost combines the expected waiting time $w^{\ell'} = \frac{h^{\ell'}}{2}$ with a fixed 5 min transfer penalty, a common value in transit route system design (Baaj and Mahmassani, 1991). Thus, each required transfer contributes $w^{\ell'} + 5$ min, consistent with the waiting-time formulation in (1) while also accounting for transfer inconvenience.

3.1.1. Virtual Node-Based Transfer Model

To model passenger routing, we construct an auxiliary directed graph $\mathcal{G}^{\text{PAX}} = (\mathcal{V}^{\text{PAX}}, \mathcal{E}^{\text{PAX}})$, referred to as the virtual-node graph; the same object is denoted $\mathcal{G}'$ in the solution algorithm of §4. The node set $\mathcal{V}^{\text{PAX}}$ comprises physical stop nodes $i \in \mathcal{V}^{\text{sub}}$ and line-specific virtual nodes v_i^ℓ for each line ℓ serving stop i. As illustrated in Fig. 2, yellow nodes denote physical stops, while colored nodes represent line-specific virtual nodes; connections between colored nodes indicate potential passenger transfers. The edge set $\mathcal{E}^{\text{PAX}}$ incorporates three distinct directed link types, each associated with a specific temporal cost. **(1) In-vehicle edges:** Connect virtual nodes v_i^ℓ and v_j^ℓ along the same line ℓ. Their weights represent in-vehicle travel times. **(2) Transfer edges:** Connect virtual nodes of different lines (e.g., v_i^ℓ

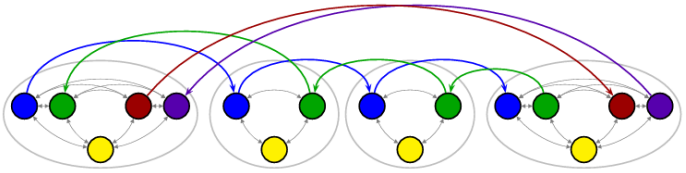


Figure 2: Example of a four-station PT network with asymmetric routes.

to $v_i^{\ell\prime}$) at the same physical node. **(3) Access& Egress edges:** Connect physical nodes to virtual nodes (e.g., i to v_i^{ℓ}). These represent initial boarding waiting times. zg~~The weights of (2) and (3) correspond to passenger waiting times, which are defined in (1) in § 3.1.~~The weight of each transfer edge in (2) equals the transfer waiting time plus the fixed 5 min transfer penalty defined above, whereas the weights of access/egress edges in (3) correspond to initial boarding waiting times; the waiting-time components are defined in (1) in § 3.1. Transfer edges are not created at origin or destination nodes.

3.2. Stochastic Modeling of Travel Times

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For the sake of realism, we model the aforementioned bus travel times t_{ij} from stop i to stop j, for each edge $(i,j) \in \mathcal{E}^{\text{sub}}$, at a given time of day, as random variables. For simplicity, we omit explicitly specifying the time of day in the notation. Let $t_{ij}^{\omega}, \omega \in \Omega$ denote a realization of this variable. This is the travel time we would measure if we take a snapshot of the network at a certain point in time. Observe that, if we take another snapshot (at the same time of day, but in another day), we would observe another travel time $t_{ij}^{\omega'}$, corresponding to another realization of variable t_{ij}. We assume that a prediction model provides predictions $\hat{t}_{ij}$ for each edge. If the prediction model were "perfect", for every sample $\omega \in \Omega$, the realizations would satisfy $\hat{t}_{ij}^{\omega} = t_{ij}^{\omega}, \forall \omega \in \Omega, (i,j) \in \mathcal{E}^{\text{sub}}$. This is obviously impossible in reality, as any realistic prediction model is subject to some errors. Let the prediction error (which is another random variable) be:

$$\epsilon_{ij} \overset{\text{def}}{=} \hat{t}_{ij} - t_{ij} \tag{2}$$

where t_{ij} and $\hat{t}_{ij}$ denote the realized and predicted travel times, respectively. The sampling procedure for this error will be presented in §5.2. Let e_{ij}^{ℓ} indicate whether edge (i,j) is used by line ℓ. The time from one terminal to the other on line ℓ is:

$$t^{\ell} = \underbrace{\sum_{i\in\mathcal{N}} \sum_{j\in\mathcal{N}} t_{ij}\, e_{ij}^{\ell}}_{\text{in-vehicle travel time}} + \underbrace{d \cdot \left(\sum_{i\in\mathcal{N}} \sum_{j\in\mathcal{N}} e_{ij}^{\ell} - 1 \right)}_{\text{dwell time at stops}}. \tag{3}$$

Here, d denotes a fixed dwell time per stop. Since the actual link travel times t_{ij} for edge (i,j) are inherently uncertain, the true line travel time t^{ℓ} cannot be determined *a priori*. We can only calculate a "predicted" $\hat{t}^{\ell}$, obtained by replacing t_{ij} with $\hat{t}_{ij}$ in (3).

3.3. Trip Travel Times

Let $\mathcal{G}^{\text{sub}}$ and $\mathcal{G}_{\text{PT}}$ denote the substrate and public transport (PT) graphs, respectively (§3.1). Each OD pair $k \in \mathcal{K}$ is defined by an origin stop $o_k \in \mathcal{V}^{\text{sub}}$ and a destination stop $d_k \in \mathcal{V}^{\text{sub}}$. We assume origins and destinations are both stops, since we take responsibility of user trips from the moment they enter PT to the moment they exit, as considered in recent graph-based public transport modeling studies (Zhao, Mihăiţă, Ou, Grzybowska and Li, 2024).

Observation 1. *Given a fixed PT network design, including line structures $(e_{ij}^{\ell}, a_i^{\ell}, b_i^{\ell})$ and fleet allocations n^{ℓ}, we evaluate the service offered to each OD pair through the minimum generalized travel time attainable on the resulting PT network. This shortest-path evaluation is consistent with standard practice in transit network design and line-planning optimization, where candidate networks are often assessed by assigning OD demand to shortest paths or by*

Symbol	Description
Sets:	
$\mathcal{N}$	Set of nodes, indexed by $i, j \in \mathcal{N}$
$\mathcal{L}$	Set of PT lines, indexed by $\ell \in \mathcal{L}$
$\mathcal{K}$	Set of origin–destination pairs, indexed by $k \in \mathcal{K}$
$\mathcal{E}$	Set of feasible edges in the network, indexed by $(i,j) \in \mathcal{E}$
Parameters:	
$q_k \in \mathbb{N},\ k \in \mathcal{K}$	Demand for OD pair k (passengers) - §3.4.2.
$t_{ij} \in \mathbb{R}_+,\ (i,j) \in \mathcal{E};\ \ t^{\omega}_{ij}, (i,j) \in \mathcal{E}, \omega \in \Omega$	t_{ij} is the true travel time on link (i,j); this is a random variables; we denote its realizations as t^{ω}_{ij} - §3.2.
$t^0_{ij} \in \mathbb{R}_+,\ (i,j) \in \mathcal{E}$	Baseline link time (the ones referred to nominal conditions) - §3.3.
$\hat{t}_{ij} \in \mathbb{R}_+,\ (i,j) \in \mathcal{E}$	Predicted travel time on link (i,j) - §3.4.2.
$t^0_k \in \mathbb{R}_+,\ k \in \mathcal{K}$	Baseline trip travel time: travel time for OD pair k in the original network design $\mathcal{G}^{\text{org}}_{\text{PT}}$ under nominal (unperturbed) conditions - §3.3.
$o_k, d_k \in \mathcal{N},\ k \in \mathcal{K}$	Origin and destination node of OD pair k - §3.3.
$e^{\ell,0}_{ij} \in \{0,1\},\ i,j \in \mathcal{N},\ \ell \in \mathcal{L}$	Original network configuration (1 if link (i,j) is used by line ℓ) - §3.4.2
$M, d \in \mathbb{R}_+$	Large constant, Fixed dwell time
Upper-level decision variables (network design):	
Binary Variables:	
$y^{\ell},\ \ell \in \mathcal{L}$	1 if line ℓ is activated (operated)
$e^{\ell}_{ij},\ i,j \in \mathcal{N},\ \ell \in \mathcal{L}$	1 if edge (i,j) is used by line ℓ in new network
$a^{\ell}_i, b^{\ell}_i,\ i \in \mathcal{N},\ \ell \in \mathcal{L}$	1 if node i is the start/end terminal of line ℓ
Discrete or Continuous Variables:	
$n^{\ell} \in \mathbb{N},\ \ell \in \mathcal{L}$	Number of vehicles on line ℓ (fleet size)
$u^{\ell}_i \in \mathbb{N},\ i \in \mathcal{N},\ \ell \in \mathcal{L}$	Auxiliary variable for subtour elimination (MTZ): it denotes the order in which stop i appears in line ℓ.
$\hat{w}^{\ell} \in \mathbb{R}_+,\ \ell \in \mathcal{L}$	Average waiting time for line ℓ
$\hat{t}^{\ell} \in \mathbb{R}_+,\ \ell \in \mathcal{L}$	Travel time from one terminal to the other on line ℓ
Induced quantities (passenger assignment outputs):	
$p^{k\ell}_{ij} \in \{0,1\}$	1 if OD pair k traverses arc (i,j) using line ℓ
$\hat{t}_k \in \mathbb{R}_+,\ k \in \mathcal{K}$	zg~~Travel time of OD pair k~~Predicted trip travel time of OD pair k in the redesigned network $\mathcal{G}^{\text{adpt}}_{\text{PT}}$ (see §3.3)
zg$\hat{t}^0_k \in \mathbb{R}_+,\ k \in \mathcal{K}$	zgPredicted trip travel time of OD pair k on the original network $\mathcal{G}^{\text{org}}_{\text{PT}}$ under predicted link times $\hat{t}_{ij}$, i.e., the predicted cost of not redesigning (see §3.3)

Table 2
Notation for PT Network Re-Design Problem

ensuring that passengers can be served on shortest routes, rather than by embedding a full behavioral route-choice model (Ahmed, Mumford and Kheiri, 2019). The computed shortest path should therefore be interpreted as the best travel-time option made available by the design, not as an operational constraint imposed on individual passengers. Users may still choose other feasible routes, but such behavioral heterogeneity is outside the scope of the present optimization model. With additive link costs, the OD-specific benchmark paths are induced by the design and can be computed via standard shortest-path algorithms.

Accordingly, when computing shortest paths we assume links are uncongested: link travel times are exogenous to passenger volumes, so the reported trip times are user-optimal times that abstract away vehicle capacity and in-vehicle crowding (a redesigned line is never penalized for the demand it attracts). This is a standard simplification at the network-design level (Goerigk and Schmidt, 2017; Schöbel and Urban, 2022; John et al., 2014), and the travel-time improvements reported in §6 should be read as uncongested user-optimal gains; capacity- and crowding-aware assignment is left to future work (§7).

For any network configuration $\mathcal{G}_{\text{PT}}$ and any realization $\omega \in \Omega$, each link (i,j) can be traversed in time t^{ω}_{ij}. Based on such values, we compute shortest paths for any OD pair $k \in \mathcal{K}$ and denote the trip travel time as t^{ω}_k. zg~~Note that t^{ω}_k includes (i) in-vehicle travel times, (ii) dwell times, (iii) initial waiting times, and (iv) transfer waiting times,~~

~~according to §3.1.1.~~Note that t_k^ω includes (i) in-vehicle travel times, (ii) dwell times, (iii) initial waiting times, and (iv) transfer waiting times plus the fixed transfer penalties, according to §3.1.1. Let t_{ij} be the random variable with realizations $t_{ij}^\omega, \omega \in \Omega$. To formalize these paths, we define binary dependent variables $p_{ij}^{k\ell,\omega}$ indicating whether arc (i,j) on line ℓ belongs to the shortest path for OD pair k in realization $\omega \in \Omega$. These variables are uniquely determined by the network state and computed algorithmically (§4) rather than being direct decision variables.

Let $\mathcal{G}_{\text{PT}}^{\text{org}}$ denote the original PT network, i.e., "master" schedule of PT over a typical day of operation. Classically, this is the network that will be operated every day. We instead allow adaptation, which means that the PT line network at a certain time is $\mathcal{G}_{PT}$, which may or may not correspond to the original one (Fig. 1). In our framework, three notions of OD travel time are defined: (i) the actual travel time t_k, described in the paragraph above; we consider its realizations when evaluating the performance of our method. (ii) the predicted travel time $\hat{t}_k$, computed as the sum of the predicted link times $\hat{t}_{ij}$ along the shortest paths described in the paragraph above; we use these predicted values as input parameters of optimization (5)-(14); (iii) the baseline travel time t_k^0, defined as the OD travel time on the original network design $\mathcal{G}_{\text{PT}}^{\text{org}}$ under baseline link times $t_{ij}^0, \forall (i,j) \in \mathcal{E}$). zgFor completeness, we additionally denote by $\hat{t}_k^0$ the predicted trip travel time of OD pair k on the original network $\mathcal{G}_{\text{PT}}^{\text{org}}$ under the predicted link times $\hat{t}_{ij}$; this represents the predicted cost of not redesigning, and is distinguished from $\hat{t}_k$, which is the predicted trip travel time on the redesigned network $\mathcal{G}_{\text{PT}}^{\text{adpt}}$.

3.4. Problem Statement

This research tackles the aaproactive re-optimization of public transport lines (e.g., one hour ahead), aa~~redesigning them only~~triggered when aa~~the forecast-trigger condition in Rule 1 is met to adapt to fluctuating and unknown road conditions~~traffic conditions heavily deviate from the nominal ones, in order to preserve user and operator performance.. The proposed optimization method has three complementary objectives: (i) minimizing user cost by reducing trip travel time deterioration due to fluctuations of substrate link times, with respect to the nominal ones (resulting in changes of actual travel time t_k experienced on OD pair k, relative to the baseline travel time experienced in the original network, which we denote as t_k^0), (ii) lowering operator costs represented by total vehicle-hours traveled by the fleet of buses, and (iii) avoiding excessive changes in the PT network compared to the original to improve operational manageability and prevent user disorientation.

Our optimization problem addresses the following three decision problems: (a) how to dynamically modify PT lines (line topology, fleet allocation, headways) to adapt to actual road conditions when they severely deviate from the expected ones? (b) which route should we make available for users to travel among OD pairs? and (c) where to locate the terminals of each line?

3.4.1. Timeframe and Stability Considerations

Reconfiguring PT lines may affect users and operators in distinct ways.

For passengers, changes in routes can reduce perceived regularity and predictability, a recognized determinant of transit network design quality (van Oort and van Nes, 2009). However, large urban public-transport systems already impose frequent on-the-fly journey adaptation; such changes are therefore not a completely new burden introduced by our approach, which at least attempts to prevent the degradation of user performance. In New York City Subway, major incidents (delaying at least 50 trains) occurred more than 500 times a year (Office of the New York State Comptroller, 2025, p. 2, "Major Incidents"; p. 3, Fig. 3); moreover, 50% of surveyed subway users reported having been forced by delays to change mode (Office of the New York City Comptroller, 2017, p. 9). Users are therefore already accustomed to experiencing public transport as an intrinsically variable system. As a consequence, navigation apps are increasingly used (Federal Highway Administration, 2016) and more and more passengers rely on them (Durand, Hamersma, van Oort and Hoogendoorn, 2024). Ghahramani and Brakewood (2016, p 147) report that 80% of New York PT passengers use those apps twice or more in one day. Li and Shalaby (2024) show that "people seek out transit information the most during times of uncertainty" to find their best itinerary. Building on this trend, we assume that, under our proposed proactive approach, smartphone navigation apps provide passengers with the best itinerary, in case traffic variability is so high that a network redesign is triggered, thereby enabling them to navigate the reconfigured line structure. In addition to the availability of smartphone apps, behavioral trends of PT users suggest that they could be willing to adapt their trips to reconfigured network structures if this would allow them to arrive at their destination more efficiently. Indeed, empirical evidence suggests that passengers operate a trade-off between efficiency and regularity (Devarasetty, Burris and Shaw, 2012). Specifically, some travelers may prefer an alternative path that ensures on-time arrival over

a habitual route that, under adverse traffic conditions on a given day, results in delay. In this sense, deviations from regular routes are not necessarily detrimental if they yield tangible gains in travel time reliability.

For the operator, frequently updated itineraries may increase cognitive load on the drivers. Nonetheless, in practice, several agencies operate services with "route deviation", in which vehicles may depart from established routes within a defined area (National Rural Transit Assistance Program (National RTAP), 2024). We assume that such operating areas have been defined ex ante for liability reasons. The operational feasibility of these assumptions—one-hour advance notice, predefined deviation zones, and app-based passenger guidance—is therefore context-dependent. They will be more realistic in settings where flexible routing will be organizationally and technologically supported (e.g., with forthcoming vehicle automation (Peled et al., 2019)), while in more rigid operating environments the results presented here should be interpreted as upper bounds on the performance gains that could be achieved as operational practices, vehicle automation, and passenger-information systems evolve. aaThe ways in which the results of our method would be used in practice could vary. One could, for instance, assume that operators could first revise the redesign issued by the proposed method, instead of allowing the line changes in an automatic manner. Or the method could be used as a decision support system for operational service adjustment, subject to the final judgment of the operators.

Although PT line reconfiguration is implementable, PT lines are more easily manageable if disruptive reconfigurations are limited. For this reason, we measure structural stability using the Jaccard index (Jaccard, 1901), defined for the edge sets of the original and redesigned networks as

$$J(\mathcal{E}_{\text{orig}}, \mathcal{E}_{\text{new}}) = \frac{|\mathcal{E}_{\text{orig}} \cap \mathcal{E}_{\text{new}}|}{|\mathcal{E}_{\text{orig}} \cup \mathcal{E}_{\text{new}}|} \in [0, 1]. \tag{4}$$

It provides a simple, system-wide measure of overlap and has been shown to be particularly effective in sparse networks (Travieso, Benatti and Costa, 2024), which reflect typical PT topologies. Unlike sequence-based metrics such as edit distance (Cheng, Yan, Wu, Huang and Zhang, 2025), which focus on local route alignment and scale poorly in multi-line systems (Jain, Roy, Meher, Chakrabarti and De, 2024), the Jaccard index offers a tractable, network-level notion of similarity that directly matches our design objectives. We assume access to a short-term traffic prediction module, as is often the case in urban operations (Li, Yu, Shahabi and Liu, 2017; Liao, Zhang, Wu, McIlwraith, Chen, Yang, Guo and Wu, 2018), and focus on the subsequent optimization step. zgAs shown in Fig. 1, we consider an hourly monitoring policy: if average relative deviation $(1/|\mathcal{E}^{\text{sub}}|)\sum_{(i,j)\in\mathcal{E}^{\text{sub}}} |\hat{t}_{ij} - t^0_{ij}|/t^0_{ij}$ remain within a tolerance (e.g., 0.1), the network is kept unchanged or reset to the original configuration to avoid accumulated drift; larger deviations trigger a redesign. This selective updating strategy preserves temporal consistency while facilitating response to fluctuating traffic conditions. Moreover, deciding reconfigurations (when needed) always 1 hour in advance, gives the operator and drivers the needed time to adapt. In the hourly implementation, this one-hour advance-notice window is used as an offline planning period before the next operating interval. Because structural continuity is carried by the objective Z_3, we report the full trade-off between structural overlap and passenger benefit along the Pareto front rather than committing to a single operating point; the monitoring tolerance and the redesign trigger introduced above are the only thresholds fixed a priori.As shown in Fig. 1, we use the following hourly trigger.

Rule 1 (Forecast-triggered line redesign)**.** *At the start of a time slot, define* $\bar{\Delta} \overset{\text{def}}{=} \frac{1}{|\mathcal{E}^{sub}|}\sum_{(i,j)\in\mathcal{E}^{sub}} \frac{|\hat{t}_{ij} - t^0_{ij}|}{t^0_{ij}}$*. For a monitoring tolerance* $\delta_{\text{trig}} \in [0, 1]$*, the announced network is kept if* $\bar{\Delta} \le \delta_{\text{trig}}$*. If* $\bar{\Delta} > \delta_{\text{trig}}$*, the operator solves* (5)–(14) *using the predicted link times* $\hat{t}_{ij}$*, and the selected solution becomes* $\mathcal{G}^{\text{adpt}}_{\text{PT}}$ *for the next one-hour slot, subject to operator approval.*

The rule determines whether a redesign is attempted. Conditional on triggering, the extent of structural change is controlled by Z_3 in (5). The Jaccard-based structural term also provides a simple stability guarantee. If each redesigned public-transport network $\mathcal{G}^{(k)}_{\text{PT}}$ remains sufficiently similar to the original configuration $\mathcal{G}^{(0)}_{\text{PT}}$, then the similarity between any two redesigns is bounded from below, regardless of how many time slots separate them.

Proposition 2 (zgHorizon-independent pairwise stability)**.** *Let* $\mathcal{E}^{(k)}$ *denote the set of active edges in the redesigned network at step k,* zg*and analogously for the original network and for the next redesign. Then, for any consecutive redesigns k and k+1and* $\mathcal{E}^{(0)}$ *for the original network. Then, for any two redesign steps k and* $k + n$ $(n \ge 1)$*, it holds*

that[1]

$$J\big(\mathcal{E}^{(k)}, \mathcal{E}^{(k+n)}\big) \;\geq\; J\big(\mathcal{E}^{(k)}, \mathcal{E}^{(0)}\big) + J\big(\mathcal{E}^{(k+n)}, \mathcal{E}^{(0)}\big) - 1.$$

Rule 1 specifies a possible deployment mechanism, whereas Proposition 2 gives an analytical stability property induced by the Jaccard-based structural term. Together, they motivate a rolling, multi-period deployment of the proposed method. The experiments in this paper, however, evaluate the redesign step itself: namely, the quality of a single forecast-conditioned redesign under varying travel-time variability. A full empirical evaluation of the rolling trigger over realized multi-hour traffic time series is left to future work.

3.4.2. Objective Functions

We formulate our public transport network adaptation problem as a three-objective optimization, as expressed in (5). The first objective, Z_1, minimizes passenger travel time deterioration, where q_k represents the travel demand flow. The use of $\max(\hat{t}_k - t_k^0, 0)$ guarantees that only travel-time increases relative to the nominal benchmark are penalized, thereby avoiding situations where improvements for certain OD pairs compensate for deterioration in others. The second objective Z_2 minimizes operational costs through total vehicle-hours. The third objective Z_3 minimizes network structural changes using the Jaccard index (§ 3.4.1), calculated as the ratio between the number of shared edges (i.e., those present both in the original and in the re-designed network) and the total number of edges (i.e., the union of the edges in the two networks). Structural continuity is handled as an objective (Z_3).

$$\min_{\mathcal{G}_{\mathrm{PT}}^{\mathrm{adpt}}} \begin{pmatrix} Z_1 \\ Z_2 \\ Z_3 \end{pmatrix} = \begin{pmatrix} \underbrace{\sum_{k\in\mathcal{K}} q_k \cdot \max(\hat{t}_k - t_k^0, 0)}_{\text{Passenger travel time deterioration}} \\ \underbrace{\sum_{\ell\in\mathcal{L}} n^\ell \cdot 2\hat{t}^\ell}_{\text{Total vehicle-hours (operating cost)}} \\ \underbrace{1 - \dfrac{\sum_{\ell\in\mathcal{L}} \sum_{(i,j)\in\mathcal{E}} \min(e_{ij}^\ell, e_{ij}^{\ell,0})}{\sum_{\ell\in\mathcal{L}} \sum_{(i,j)\in\mathcal{E}} \max(e_{ij}^\ell, e_{ij}^{\ell,0})}}_{\text{Route dissimilarity}} \end{pmatrix} \tag{5}$$

In (5), $e_{ij}^{\ell,0}$ and e_{ij}^ℓ denote whether line ℓ uses substrate link (i,j) in the original and the redesigned network, respectively. Observe that $e_{ij}^{\ell,0}$ are parameters of our problem, whereas e_{ij}^ℓ are decision variables. Observe that when solving our optimization problem, the realized link travel times t_{ij} are not available, we will thus optimize the decision variables based on prediction $\hat{t}_{ij}$ and then evaluate, ex-post in the results, the objective function evaluated using the realized link travel times t_{ij}.

zg**Variable domains.** Binary: $y^\ell, e_{ij}^\ell, a_i^\ell, b_i^\ell \in \{0,1\}$. Integer: $n^\ell \in \mathbb{Z}_{\geq 1}$, $u_i^\ell \in \{1,\dots,|\mathcal{N}|\}$. Continuous: $\hat{w}^\ell, \hat{t}^\ell \in \mathbb{R}_{\geq 0}$. The induced OD travel time $\hat{t}_k$ is a function of the design variables and the predicted link times $\hat{t}_{ij}$; it is computed algorithmically (see §4.3) and is not a free variable of the program.

3.4.3. Constraints

We now report the constraints that, together with the objective (5), formalize the upper-level network design problem. Passenger route variables are treated as induced (follower) variables and are discussed separately.

Performance-Related Constraints (1) and (3).

Line Structure Constraints

$$\sum_{i\in\mathcal{N}} a_i^\ell = y^\ell, \qquad \sum_{i\in\mathcal{N}} b_i^\ell = y^\ell \qquad\qquad \forall \ell \in \mathcal{L} \tag{6}$$

[1] The proof is provided in App. A.

$$a_i^\ell + b_i^\ell \le 1 \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{7}$$

$$e_{ij}^\ell + e_{ji}^\ell \le y^\ell \qquad \forall \{i,j\} \in \mathcal{E},\ \forall \ell \in \mathcal{L} \tag{8}$$

$$\sum_{j:\{i,j\}\in\mathcal{E}} e_{ji}^\ell + a_i^\ell = \sum_{j:\{i,j\}\in\mathcal{E}} e_{ij}^\ell + b_i^\ell \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{9}$$

$$\sum_{j:\{i,j\}\in\mathcal{E}} e_{ij}^\ell \le 1, \qquad \sum_{j:\{i,j\}\in\mathcal{E}} e_{ji}^\ell \le 1 \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{10}$$

$$u_i^\ell - u_j^\ell + 1 \le |\mathcal{N}|\left(1 - e_{ij}^\ell\right) \qquad \forall \{i,j\} \in \mathcal{E},\ \forall \ell \in \mathcal{L} \tag{11}$$

$$1 \le u_i^\ell \le |\mathcal{N}| \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{12}$$

$$n^\ell \ge y^\ell, \qquad n^\ell \le \bar{n}\, y^\ell \qquad \forall \ell \in \mathcal{L} \tag{13}$$

$$\sum_{\ell\in\mathcal{L}} n^\ell \le V_{\max} \tag{14}$$

Constraints (6)–(14) establish the topological structure and operational limits of the designed network. Constraints (6)–(7) synchronize line activation with terminal selection, mandating that every active line ($y^\ell = 1$) connects exactly one initial terminal to one final terminal, thereby preventing zero-length loops. Constraint (8) ensures that if a line is not activated ($y^\ell = 0$), there are no links associated to it. Otherwise, it ensures that the line is constructed in one direction (links (i,j) and (j,i) cannot be both selected) and then buses go back and forth along it (as specified in page 5). Route continuity and linearity are enforced by the flow conservation in (9) combined with the degree Constraints in (10); the latter restrict nodes to a maximum of one predecessor and one successor, strictly imposing a non-branching simple path topology. Acyclic integrity is further guaranteed by Constraints (11)–(12), which adapt the Miller-Tucker-Zemlin (MTZ) formulation to eliminate disconnected subtours that might otherwise satisfy flow balance: if link (i,j) is used in line ℓ, then node j should appear after i along the line (its order of appearance (u_j) must be higher than i). Finally, resource allocation is governed by Constraints (13)–(14), which assign vehicles only to active lines and enforce the global fleet budget.

Induced Passenger Paths. Passenger routes are not decision variables of the network design problem. Given a fixed network design $(\mathbf{e}, \mathbf{n})$, passenger paths for each OD pair $k \in \mathcal{K}$ are deterministically induced by solving a shortest-path problem on the auxiliary passenger graph $\mathcal{G}^{\mathrm{PAX}}$ defined in §3.1.1. Let $\hat{t}_k$ denote the resulting shortest-path travel time under predicted link costs $\hat{t}_{ij}$. For bookkeeping and interpretation, we also define $p_{ij}^{k\ell} \in \{0,1\}$ as an induced indicator equal to one if the shortest path for OD pair k traverses arc (i,j) on line ℓ, and zero otherwise. Accordingly, $\hat{t}_k$ can be expressed as the sum of the costs of the traversed arcs (including in-vehicle, dwell, access, and transfer components) on $\mathcal{G}^{\mathrm{PAX}}$. These induced quantities are computed algorithmically and are not part of the optimization decision set.

4. Solution Algorithm

The optimization model presented in § 3 is a mixed-integer formulation featuring discrete network design choices and nonlinear objective terms. While the delay penalty in Z_1 is piecewise linear, the fractional Jaccard objective Z_3 introduces nonconvexity and global variable coupling. Since an exact linearization of this ratio would require auxiliary constraints that severely compromise scalability, we retain the fractional structure. Given the tri-objective nature of the problem and the nonconvex trade-offs, we adopt an NSGA-III framework (Algorithm 1) to approximate the Pareto front. As illustrated in Fig. 3, we adopt a **design–evaluation decomposition** for computational tractability. The evolutionary search operates on network design variables (line topologies e_{ij}^ℓ and fleet allocations n^ℓ). For any fixed design $(\mathbf{e}, \mathbf{n})$, passenger routing is not treated as an independent optimization model; instead, it is a deterministic **decode-and-assign mapping** induced by Observation 1. Specifically, OD travel times and the corresponding assignment outputs are computed by shortest-path calculations on the virtual graph G' (§3.1.1), which provides the objective values (Z_1, Z_2, Z_3) used by NSGA-III. Accordingly, the proposed workflow is structured into three coordinated modules: (i) **Solution Encoding and Initialization** (§ 4.1) constructs a population of structurally feasible designs; (ii) the **NSGA-III Evolutionary Loop** (§ 4.2) drives the feasible solution search, employing reference-point-based selection and variation & repair operators to evolve the topology; and (iii) the **Decode-and-Assign Strategy** (§ 4.3) performs deterministic passenger assignment to compute the objective vector (Z_1, Z_2, Z_3).

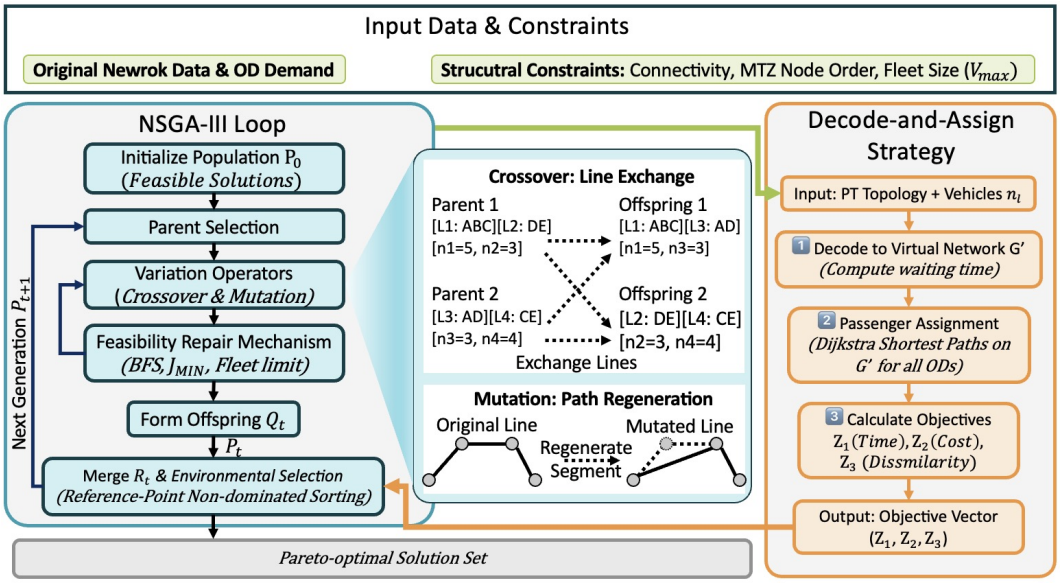


Figure 3: Flowchart of the Proposed NSGA-III-based optimization method.

Algorithm 1 NSGA-III based Adaptive PT Network Optimization

Input: $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, $\mathcal{K}$, $\mathcal{L}$, $V_{\max}$, q_k, $\hat{t}_{ij}$, *PopSize*, *MaxIter*, N_{ref}, P_c, P_m, $J_{\min}$
Output: Approximate Pareto set $\mathcal{P}^*$

Initialize a population $\mathcal{P}$ of size *PopSize* via §4.1 and §4.2.1 (using parameter $J_{\min}$)
Evaluate (Z_1, Z_2, Z_3) for all $p \in \mathcal{P}$ via decode-and-assign (§4.3)
for $iter = 1$ **to** *MaxIter* **do**
    Select parents from $\mathcal{P}$
    Generate offspring $\mathcal{Q}$ by crossover (§4.2.2) with P_c and mutation (§4.2.3) with P_m
    Apply feasibility-oriented repair to offspring when line-level structural infeasibility (e.g., disconnection or discontinuity) is detected (Alg. 2)
    Evaluate (Z_1, Z_2, Z_3) for all $q \in \mathcal{Q}$ via decode-and-assign (§4.3)
    $\mathcal{R} \leftarrow \mathcal{P} \cup \mathcal{Q}$
    Perform non-dominated sorting on $\mathcal{R}$
    Apply NSGA-III selection with N_{ref} reference points to obtain next $\mathcal{P}$ (§4.2.4)
end for
return the non-dominated solutions in $\mathcal{P}$ as $\mathcal{P}^*$

4.1. Solution Encoding

In Alg. 1, each PT line ℓ is represented as an ordered simple path $p_\ell = (i_1, i_2, \ldots, i_{n_\ell})$ with $i_j \in \mathcal{N}$ and no repeated nodes, which enforces acyclicity by construction according to constraints (6)–(14). PT lines may either be constructed from *scratch* or obtained by *reconstructing routes from the existing network*. Starting from a current node i, the algorithm maintains a visited-node set $\mathcal{V}^{\ell}_{\text{visited}}$ and defines the set of feasible candidates as $\mathcal{N}_{\text{unvisited}}(i) = \{ j \in \mathcal{N}(i) : j \notin \mathcal{V}^{\ell}_{\text{visited}} \}$, which is used whenever a new path segment needs to be generated. For lines constructed from scratch, the path is iteratively extended by selecting the next node $j \in \mathcal{N}_{\text{unvisited}}(i)$ according to the travel-time weighted probability

$$\mathbb{P}(j \mid i) = \frac{1/t_{ij}}{\sum_{k \in \mathcal{N}_{\text{unvisited}}(i)} 1/t_{ik}}, \tag{15}$$

which favors temporally efficient connections while preserving stochastic exploration. When a line is derived from the existing network, local modifications are applied by replacing a selected route segment with an alternative feasible path that preserves the original endpoints.

Simultaneously, vehicle allocation is managed by assigning integer variables n^ℓ to each line. During initialization (Alg. 1, line 1), each line is pre-allocated one vehicle to guarantee basic service availability. The remaining $V_{\max} - |\mathcal{L}|$ vehicles are then distributed across lines proportionally to their initial one-way travel times t^ℓ (computed based on nominal link travel times t_{ij}).

4.2. Evolutionary Search Procedure

Each individual encodes a complete PT network design, comprising a set of line path representations and the associated vehicle allocations. Starting from an initial population, the algorithm evolves solutions in a generational manner. At each generation, offspring are produced through selection, variation, and evaluation, followed by the NSGA-III reference-point-based *environmental selection*, which prioritizes non-dominated solutions while maintaining diversity along the Pareto front. The overall procedure is summarized in Alg. 1.

4.2.1. Population Initialization

During population initialization (Alg. 1, line 1), each individual is generated using the line construction rules described in § 4.1. PT lines are either constructed from scratch or obtained by locally perturbing routes reconstructed from the existing network. Given a set of lines, vehicle allocations are initialized by assigning one vehicle to each active line to guarantee basic service availability. The remaining fleet is then distributed across lines according to the chromosome encoding scheme described in § 4.1, ensuring $\sum_{\ell\in\mathcal{L}} n^\ell = V_{\max}$.

To expedite initialization, candidate solutions are screened via a **connectivity check** on the undirected graph induced by $\mathcal{E}_{\text{PT}} = \bigcup_{\ell\in\mathcal{L}} \mathcal{E}_\ell$, i.e., the union of all active line edges in $\mathcal{G}_{\text{PT}}$. By employing a Union–Find data structure, we efficiently identify connected components, and verify that, for every OD pair $k \in \mathcal{K}$, the origin o_k and destination d_k belong to the same connected component (which implies that it is possible to go from one to the other). The procedure is repeated until *PopSize* individuals (i.e., PT network structures) are generated that satisfy: (i) structural validity (lines are simple-paths, with no branches nor cycles), (ii) connectivity-based serviceability, i.e., all OD pairs $k \in \mathcal{K}$ are connected within $\mathcal{G}_{\text{PT}}$, and (iii) similarity, i.e., individual $\mathcal{G}_{\text{PT}}$ is not extremely different from original PT network $\mathcal{G}_{\text{PT}}^{\text{org}}$, i.e., the Jaccard similarity index (4) must be above threshold $J_{\min}$. These criteria enforce a necessary (but not sufficient) condition for OD reachability; full feasibility and network performance are subsequently evaluated using the decode-and-assign objective.

4.2.2. Crossover Operation

With probability P_c, a crossover operator is applied to two parent solutions, denoted as parents A and B. Each solution consists of a fixed set of PT lines, where each line is treated as a basic inheritance unit characterized jointly by its route topology and vehicle allocation. The crossover exchanges complete lines between parents as the primary recombination mechanism; any subsequent modifications are limited to feasibility restoration.

Line inheritance is governed by a binary selection mask $\boldsymbol{\xi} = (\xi_\ell)_{\ell\in\mathcal{L}}$, where each component is independently drawn as $\xi_\ell \sim \text{Bernoulli}(0.5)$, so that both parents are treated symmetrically and structural diversity is promoted at the line level. Given $\boldsymbol{\xi}$, the offspring design is defined by block-wise inheritance:

$$(e_{ij}^\ell)^{\text{off}} = \xi_\ell (e_{ij}^\ell)^A + (1-\xi_\ell)(e_{ij}^\ell)^B, \qquad \forall \ell \in \mathcal{L},\ (i,j) \in \mathcal{E}, \tag{16}$$

$$(n^\ell)^{\text{off}} = \xi_\ell (n^\ell)^A + (1-\xi_\ell)(n^\ell)^B, \qquad \forall \ell \in \mathcal{L}. \tag{17}$$

Whenever a line topology is inherited from a parent, its vehicle allocation is inherited jointly, thereby preserving the coupling between route geometry and service frequency.

If any inherited line becomes structurally infeasible (e.g., disconnected or discontinuous), a feasibility-oriented restoration operator is applied to recover a valid simple-path representation (Alg. 2). Finally, fleet constraints ($n^\ell \geq 1$ for all active lines and $\sum_{\ell\in\mathcal{L}} n^\ell = V_{\max}$) are enforced via minimal ± 1 adjustments on a small subset of lines. This correction is a feasibility projection and is not guided by objective improvement.

4.2.3. Mutation Operation

With probability P_m, a mutation perturbs a single line ℓ via local structural modifications, including node insertion, node deletion, or segment reconnection. These operations introduce stochastic variations in line topology while preserving the overall chromosome structure. If the mutation results in a disconnected or discontinuous line, feasibility is restored by reconnecting the affected segments (Alg. 2). This repair step is feasibility-oriented and does not explicitly

consider objective values. Finally, global fleet conservation ($\sum_{\ell \in \mathcal{L}} n^{\ell} = V_{\max}$) is enforced in the same manner as described in § 4.2.2.

4.2.4. Environmental Selection

After offspring generation, parent and offspring populations are merged and ranked via non-dominated sorting. Environmental selection follows the standard NSGA-III mechanism, employing a set of reference points N_{ref} to maintain population diversity (Deb and Jain, 2014).[2]

4.3. Objective Evaluation via Decode-and-Assign

The evaluation phase (Alg. 1, line 7) maps each chromosome into the objective space through a systematic decode-and-assign procedure. Following Observation 1, we treat passenger routing as a deterministic response to the network configuration, thereby decoupling routing behavior from the upper-level network design decisions. For each individual, the algorithm first constructs a comprehensive virtual-node graph $\mathcal{G}'$ that integrates in-vehicle travel times and waiting times w^{ℓ}, the latter being derived from the allocated vehicle units n^{ℓ} (§ 3.1.1). To evaluate the travel-time deterioration Z_1, the algorithm solves a shortest-path problem (Dijkstra) for every OD pair $(o_k, d_k) \in \mathcal{K}$ on $\mathcal{G}'$. The resulting shortest path yields the predicted OD travel time $\hat{t}_k$. Feasibility is enforced by construction: $\mathcal{G}'$ includes only arcs corresponding to active line segments in the decoded network design, and OD paths are initialized through access/egress connections between physical stops and their associated line-specific virtual nodes (§3.1.1). This shortest-path evaluation is performed for each candidate network, consistently reflecting the current decoded topology and headways. Simultaneously, the operational cost Z_2 and network dissimilarity Z_3 are computed. Specifically, Z_3 is quantified using a line-specific Jaccard similarity, where each network element is defined as a triplet $(\min(i,j), \max(i,j), \ell)$. This representation ensures that identical physical links served by different lines are treated as distinct entities, accurately capturing the topological shifts in the service network.

5. Evaluation Method

In order to test if our zg~~real-time~~forecast-triggered PT re-design method would perform well in a realistic situation with realistic uncertain fluctuations of link times, we need to tackle the following two challenges: (i) link travel times are inherently stochastic, and (ii) re-design decisions are based on travel time predictions, which might deviate from the true ones. To account for (i), we generate stochastic travel times for all links, with different levels of variability (§ 5.1). zg~~To account for (ii), we then generate, in § 5.2, prediction errors $e_{ij}, \forall (i,j) \in \mathcal{E}^{\text{sub}}$, statistically representative of the level of inaccuracy of well established prediction models (we consider Diffusion Convolutional Recurrent Neural Network (DCRNN) - see §5.2).~~To account for (ii), we generate empirical log-relative prediction errors $\eta_{ij}, \forall (i,j) \in \mathcal{E}^{\text{sub}}$, statistically representative of the inaccuracy of a well-established prediction model (we use the Diffusion Convolutional Recurrent Neural Network (DCRNN); see §5.2 and App. B.3).

5.1. Model of Travel Time

To account for the variability in travel times, we model them as random variables, according to § 3.2. The shape we choose for this random variable is a log-normal distribution, which effectively captures the skewed nature of travel time data (Guessous, Aron, Bhouri and Cohen, 2014) and accounts for extreme delay scenarios that can occur in reality, although not often. On each link (i,j), we generate travel time realization t_{ij}^{ω} by performing Monte-Carlo sampling from random variable $t_{ij} \sim \text{log-normal}(\mu_{ij}, \sigma_{ij}^2)$ such that the median of t_{ij} corresponds to the nominal time reported in the Mandl's dataset, and the coefficient of variation (cv) matches a predetermined value: we test $cv = 0.05, 0.25, 0.5, 1.0$. The parameterization used in this sampling is stated explicitly below.

Proposition 3 (Log-normal parameterization for prescribed median and CV)**.** *Let t_{ij} be a lognormal random variable representing the travel time on link $(i,j) \in \mathcal{E}$ with median m_{ij} and coefficient of variation cv. The parameters (μ_{ij}, σ_{ij}) of the underlying normal distribution satisfy*

$$\mu_{ij} = \ln(m_{ij}), \qquad \sigma_{ij} = \sqrt{\ln(1 + cv^2)}.$$

To test the performance of our optimization when faced with different levels of hypothetical traffic variability, for each cv, we generate a set of samples Ω_{cv}, where each sample $\omega \in \Omega_{cv}$ corresponds to a collection of travel times $\{t_{ij}^{cv,\omega} \mid (i,j) \in \mathcal{E}^{\text{sub}}\}$ across the network. The proof of Proposition 3 is provided in App. B.1.

[2] The implementation details are provided in App. C.3.

5.2. Travel Time Prediction zg~~with KDE-Based Error Modelling~~via Empirical Log-Relative Error Sampling

Our optimization uses predicted link travel times $\hat{t}_{ij}$, while actual link travel times t_{ij} are unknown at decision time. We therefore evaluate whether the generated line plans remain effective even in presence of inevitable prediction errors. Compliant with most traffic forecast models, we predict speeds on links, rather than directly predicting travel times. We express forecast errors as dimensionless quantities, valid both for travel times and speeds. This is why, in the following, we will just refer to "errors". In particular, let the logarithmic prediction errors be

$$\eta_{ij} \overset{\text{def}}{=} \log(\hat{t}_{ij}) - \log(t_{ij}) = \log(\hat{t}_{ij}/t_{ij}) = \log\left((d_{ij}/\hat{v}_{ij})/(d_{ij}/v_{ij})\right) = \log(v_{ij}/\hat{v}_{ij}) = \log(v_{ij}) - \log(\hat{v}_{ij}) \tag{18}$$

where v_{ij} and $\hat{v}_{ij}$ are the true and predicted speed on link ij, respectively, and d_{ij} is the length of link ij.

We can express the corresponding additive error (as defined in (2)) as $\epsilon_{ij} = \hat{t}_{ij} - t_{ij} = t_{ij}(\exp(\eta_{ij}) - 1)$.

We generate samples of prediction errors through a three-stage procedure:

- In Stage 1 (see § 5.2.1), we train a prediction model to forecast traffic speeds.
- In Stage 2 (see § 5.2.1), we use the trained model to produce one-hour-ahead speed forecasts and collect empirical logarithmic errors η_{ij}.
- zg ~~In Stage 3 (see § 5.2.2), we estimate the error probability distribution using the collected error samples via Kernel Density Estimation (KDE).~~ In Stage 3 (see § 5.2.2), we sample with replacement from the set of empirical logarithmic errors and use those samples to construct noisy travel-time forecasts.

zg~~As for the prediction model in Stage 1, we adopted DCRNN as a well-established spatio-temporal model; the goal was to obtain realistic prediction errors rather than compare forecasting models, and the adoption of DCRNN was presented as a conservative evaluation.~~We use the Diffusion Convolutional Recurrent Neural Network (DCRNN) (Li et al., 2017) as a representative spatio-temporal predictor to generate empirical one-hour-ahead prediction errors. The objective is not to benchmark forecasting models, but to expose the line-plan generation method to realistic forecast errors. The dataset-specific training and error-extraction procedure is summarized below and detailed in App. B.2.

zgThis prediction error-based construction is useful because downstream decisions may be sensitive not only to the average magnitude of forecast errors, but also to their distributional shape (Peled et al., 2021).

5.2.1. Stage 1 and 2: Prediction model and empirical prediction error extraction

We train a Diffusion Convolutional Recurrent Neural Network (DCRNN). Due to the absence of historical traffic data for Mandl's Swiss network, the DCRNN is trained on the METR-LA dataset (Li et al., 2017), which contains traffic speed measurements from 207 sensors in Los Angeles County collected at 5-minute intervals over four months. Although METR-LA is a different network from Mandl's, we apply the error patterns derived via DCRNN in the METR-LA to our Mandl's network, as a practical way to reproduce the realistic errors of predictive models. This empirical setup does not affect the methodological validity of the presented approach, which focuses on optimization under uncertainty rather than on predicting models' accuracy. For the real-world Beijing case study, instead, we utilize the Q-Traffic dataset (Liao et al., 2018), which provides traffic speed and road network data for the 6th Ring Road in Beijing, China.

zg~~Post-training, we collect the 1-hour-ahead speed predictions in each link $(i, j) \in \mathcal{E}$ and the corresponding errors ϵ_{ij}.~~Post-training, we collect the 1-hour-ahead speed predictions in each link $(i, j) \in \mathcal{E}$ and collect the resulting errors, as detailed in in App. B.2.

5.2.2. zg~~Stage 3: KDE modeling of prediction errors~~Stage 3: Empirical log-relative prediction error sampling

zg~~Following the previous framework, we modelled travel-time prediction errors as stochastic inputs to the network optimization problem. Earlier text contrasted pre-specified parametric noise distributions with a data-driven, non-parametric KDE approach.~~Following Peled et al. (2021), we treat prediction errors as stochastic inputs to the network optimization problem. Rather than prescribing a parametric distribution, or fitting a KDE to absolute prediction errors, we collect a set of empirical prediction errors extracted from DCRNN test-set predictions. Sampling these errors directly preserves the trained predictor's empirical bias, skewness, and tail behavior without imposing a Gaussian, Weibull, or other smooth distributional shape. The error construction and sampling procedure are detailed in App. B.3.

zg~~In particular, instead of assuming a Gaussian or Weibull prediction error shape, we obtained prediction errors from an~~

~~actual predicting model and fitted their distribution using KDE; the appendix detailed the implementation and reported that KDE performed best.~~The prediction errors are not recentred, so any mild empirical bias from the trained model is retained. Errors are not differentiated based on the links where they were calculated; consequently, we do not represent link-specific heterogeneity or spatio-temporal correlation.

zgTo assess the sensitivity of our method to the magnitude of the prediction errors, in our optimization we set predicted travel times for a certain realization $\omega \in \Omega$ as

$$\hat{t}_{ij}^{\omega}(\alpha) = t_{ij}^{\omega} \exp\left(\alpha\, \eta_{ij}^{\omega}\right), \tag{19}$$

where η_{ij}^{ω} is sampled from the set of empirical logarithmic prediction errors. Equivalently, $\epsilon_{ij}^{\omega} = t_{ij}^{\omega}(\exp(\alpha\eta_{ij}^{\omega}) - 1)$. The parameter α scales the empirical forecast-error magnitude: $\alpha = 1$ corresponds to the prediction error level extracted from the trained DCRNN, $\alpha = 0$ gives the perfect-foresight benchmark, and $\alpha > 1$ is used only in the forecast-sensitivity stress tests.

5.2.3. *Simulating predicted travel times*

zg~~For each simulation realization $\omega \in \Omega$ and link $(i, j) \in \mathcal{E}^{\text{sub}}$, we implement a two-step process: (1) sample actual travel time t_{ij}^{ω} from the log-normal distribution of § 5.1, and (2) sample a log-relative prediction error η_{ij}^{ω} via the empirical error sampling procedure detailed in App. B.3. The resulting predicted travel time is:~~ For each simulation realization $\omega \in \Omega$ and link $(i, j) \in \mathcal{E}^{\text{sub}}$, we implement a two-step process: (1) sample actual travel time t_{ij}^{ω} from the log-normal distribution of § 5.1, and (2) generate the additive prediction error ϵ_{ij}^{ω} via the empirical log-relative error sampling procedure detailed in §5.2.2. The resulting predicted travel time is:

$$\hat{t}_{ij}^{\omega} = \underbrace{t_{ij}^{\omega}}_{\S\ 5.1} + \underbrace{\epsilon_{ij}^{\omega}}_{\S\ 5.2}, \tag{20}$$

zgIn the main experiments we set $\alpha = 1$ in Eq. (19); alternative values are used only in the forecast-error sensitivity analysis (App. D.1).

6. Numerical Results

zg~~We evaluate the proposed adaptive public transport (PT) redesign framework on both benchmark and real-world networks.~~We evaluate the proposed framework on Mandl's benchmark network (§6.2) and a real-world Beijing network (§6.3). In all experiments, the optimizer receives predicted link times $\hat{t}_{ij}$ and produces a redesigned network; performance is then scored against the realized travel times t_{ij}^{ω}, so that reported figures reflect decision quality under imperfect forecasts. aa~~Each variability level cv represents the forecast-indicated stress of~~The following results concern a single decision slot; accordingly, the experiments assess the single-slot redesign rather than the multi-period rolling trigger of Rule. 1.

6.1. Experimental setup and metrics

Mandl's Swiss network, initially used by Mandl (1980), consisting of 15 nodes, 21 links, and a symmetric demand of 15,570 passenger trips, is employed as a canonical benchmark for comparison with existing PT network design methods (Liang, Xu and Wang, 2025; Ahern et al., 2022). To further assess the method's applicability under realistic operating conditions, we consider a real-world urban public transport network in a region in Beijing, China, recently analyzed by Wu et al. (2024), which comprises 301 bus stops, 3,758 road sections, and 29 bus lines. Its origin–destination demand matrix is constructed from smart card data collected during peak hours by aggregating trips between boarding station i and alighting station j.

Key parameters are set as follows: a fixed dwell time of $d = 1$ min per stop is assumed following Chen et al. (2022). For Mandl's network, the maximum fleet size ($V_{\max}$) is capped at 40, following the service-frequency setting of Ahern et al. (2022) to maintain comparable headways. For the Beijing network, $V_{\max}$ is set to 150 to accommodate a constant 12-minute headway across this large PT system.

6.1.1. *Compared networks and baselines*

We wish to show the benefits of adapting PT network design zg~~in real-time~~for the hour ahead, and only when the forecast-trigger condition in Rule 1 is met, to varying road conditions via our method, zg~~against current network design~~

~~approaches, which all keep the PT structure unchanged~~against selected static baselines, which keep the original line structure fixed.

For Mandl's Swiss network, the current network design approaches are represented by zg~~two networks~~the following comparison networks:

- A static integrated route-frequency design $\mathcal{G}_{\text{PT}}^{\text{static}}$, following Ahern et al. (2022), which jointly optimizes passenger travel time and operational cost.
- A state-of-the-art network design for trip travel-time minimization, $\mathcal{G}_{\text{PT}}^{\text{ver}}$, derived from the exact formulation of Vermeir, Engelen, Philips and Vansteenwegen (2021).
- zgA frequency-only adjustment baseline, denoted by $\mathcal{G}_{\text{PT}}^{\text{freq}}$, in which route structures are fixed and only service frequencies are adjusted.

For the real-world Beijing network, instead, the current design approaches are represented by

- zg~~The original bus line network, denoted by $\mathcal{G}_{\text{PT}}^{\text{orig}}$, serving as the current operational standard against which the adaptive redesign is assessed.~~The original bus line network, denoted by $\mathcal{G}_{\text{PT}}^{\text{orig}}$, serving as the current operational standard against which the adaptive redesign is assessed.
- zgA stop-skipping baseline, denoted by $\mathcal{G}_{\text{PT}}^{\text{skip}}$, in which buses skip major congested zg~~stops~~links while the underlying route structure is otherwise retained.

zg~~Across all experiments, the baseline travel time for an OD pair $k \in \mathcal{K}$, denoted by t_k^0 (§3.3), is computed as the shortest-path travel time between origin o_k and destination d_k in either network $\mathcal{G}_{\text{PT}}^{\text{static}}$, $\mathcal{G}_{\text{PT}}^{\text{ver}}$, or $\mathcal{G}_{\text{PT}}^{\text{orig}}$, depending on which one will be used as reference for comparison.~~Across all experiments, the baseline travel time for an OD pair $k \in \mathcal{K}$, denoted by t_k^0 (§3.3), is computed as the shortest-path travel time between origin o_k and destination d_k in the relevant comparison network: $\mathcal{G}_{\text{PT}}^{\text{static}}$, $\mathcal{G}_{\text{PT}}^{\text{ver}}$, $\mathcal{G}_{\text{PT}}^{\text{orig}}$, $\mathcal{G}_{\text{PT}}^{\text{freq}}$, or $\mathcal{G}_{\text{PT}}^{\text{skip}}$.

aaNote that, overall, we use two dynamic benchmarks, i.e., (i) frequency-only adjustment and (ii) stop-skipping. We only keep in each scenario the one that performed the best (namely (i) in Mandl and (ii) in Beijing). All experiments are conducted on a server equipped with two Intel Xeon Gold 5220R CPUs (48 cores, 96 threads, up to 4.00 GHz) with a maximum computational time of 1h per instance, which we use here as an upper bound for our offline computational experiments within the hourly decision cycle (see Fig. 1 and §3.4.1).[3] *(To respect double-blind review requirements, all problem instances and source code will be released in a public repository upon acceptance.)*

6.1.2. Scenarios and evaluation protocol

We assess the effectiveness of our method under varying degrees of travel time uncertainty through four coefficient of variation levels:

- Minimal Variation ($cv = 0.05$)
- Low Variation ($cv = 0.25$)
- Moderate Variation ($cv = 0.50$)
- High Variation ($cv = 1.00$).

We evaluate the method across 30 realizations of link travel times and prediction errors, using 30 random seeds (each corresponding to a single $\omega \in \Omega$ - §3.2), and quantify the resulting scenario-to-scenario uncertainty with standard deviations computed as described in Appendix D. Following our predict-then-optimize framework, the optimization phase utilizes predicted travel times $\hat{t}_{ij}$ to design adaptive networks, while the evaluation phase assesses actual performance using realized travel times $t_{ij}^{\omega}, \omega \in \Omega$. This separation enables us to quantify the impact of the prediction error on the quality of network adaptation.

[3] The values of the parameters of the proposed Alg. 1, as well as a thorough performance comparison against other multiobjective optimization methods, are extensively discussed in App. C.

No. of lines	Metric	CV=0	CV=0.05	CV=0.25	CV=0.50	CV=1.00
4	20% improv. share (IS) (%)	5.79 ± 2.99	8.25 ± 4.65	15.40 ± 5.66	24.63 ± 7.91	30.57 ± 11.60
	Op. Cost Red. (%)	10.20 ± 5.31	13.28 ± 7.37	16.86 ± 6.74	24.06 ± 12.01	39.20 ± 12.94
	Route overlap (%)	84.34 ± 5.44	83.79 ± 7.73	78.83 ± 6.65	77.53 ± 5.27	77.26 ± 5.19
6	20% improv. share (IS) (%)	8.96 ± 4.47	9.53 ± 3.60	16.31 ± 5.10	23.46 ± 7.84	31.50 ± 10.46
	Op. Cost Red. (%)	25.94 ± 4.71	28.94 ± 5.40	32.10 ± 5.69	36.75 ± 9.55	48.90 ± 10.19
	Route overlap (%)	79.58 ± 2.29	81.20 ± 2.80	78.16 ± 3.94	79.63 ± 4.47	79.16 ± 4.08
8	20% improv. share (IS) (%)	11.52 ± 2.67	13.99 ± 3.90	18.62 ± 5.64	26.84 ± 8.76	35.71 ± 11.53
	Op. Cost Red. (%)	26.97 ± 4.45	29.92 ± 3.84	31.51 ± 5.27	35.35 ± 7.98	45.35 ± 9.23
	Route overlap (%)	83.58 ± 4.07	84.95 ± 3.07	82.51 ± 4.03	80.80 ± 5.25	81.14 ± 4.23
10	20% improv. share (IS) (%)	8.83 ± 3.15	9.74 ± 4.54	14.27 ± 5.61	21.46 ± 8.88	31.21 ± 11.56
	Op. Cost Red. (%)	33.62 ± 4.18	34.65 ± 4.93	37.23 ± 5.53	40.62 ± 8.71	52.59 ± 9.17
	Route overlap (%)	77.57 ± 3.40	78.14 ± 2.76	76.96 ± 3.22	78.77 ± 4.09	78.34 ± 3.46
12	20% improv. share (IS) (%)	9.76 ± 3.57	10.34 ± 3.19	14.74 ± 3.51	21.44 ± 6.96	32.44 ± 11.91
	Op. Cost Red. (%)	35.94 ± 4.95	37.51 ± 5.54	39.56 ± 5.18	44.20 ± 6.90	54.27 ± 9.00
	Route overlap (%)	80.69 ± 2.98	80.95 ± 3.40	80.61 ± 2.78	81.46 ± 3.27	80.77 ± 4.11

No. of lines: number of bus lines; CV: coefficient of variation. 20% improvement share ($IS^{20\%}$): fraction of OD pairs whose travel- time improvement is at least 20%; Op. Cost Red. (OCR): Operational Cost Reduction (reduction in total vehicle-hours; see §3.4.2). The static route-frequency solutions for each number of bus line are taken from Ahern et al. (2022); entries report mean ± standard deviation over 30 realizations.

Table 3
Performance comparison of the solutions from Ahern et al. (2022) under different variability levels.

6.1.3. Performance metrics

To evaluate the adaptive network under uncertain travel times, we report three application-specific performance indicators aligned with the objectives in (5):

- **20% improv. share (%) (IS_{20}):** The percentage of OD pairs that achieve a travel time reduction of at least 20% compared to the original PT network.
- **Operational Cost Reduction (OCR):** percentage reduction in total vehicle-hours operated relative to the original PT network (the operational-cost proxy defined in §3.4.2).
- **Route Overlap (RO):** Jaccard-based percentage overlap (4) between the edge sets of the original and adaptive PT networks.

All three metrics depend on realized travel times and are therefore random variables; in the numerical results, we report their empirical means (and standard deviations) over 30 independent realizations.[4] zgWe distinguish the objective from the reporting metric: the optimization objective Z_1 is demand-weighted by q_k (§3.4.2), whereas mtric IS_{20} counts OD pairs without demand weighting and therefore answers "for how many OD relations does service improve," not "how many passenger-trips benefit." The aggregate, demand-weighted passenger benefit is captured separately by the Avg. Travel Time Improvement metric reported for the Beijing network (§6.3, Table 5), so that both the optimization and the reported benefit reflect the distributional, demand-driven question of which travellers gain from the redesign (Fan and Machemehl, 2011; Pereira, Schwanen and Banister, 2017).

6.2. Main performance results on Mandl's benchmark

zg~~We organize the Mandl results in three steps. First, we report aggregate performance across different variability levels and bus-line counts. We then compare full route redesign with a frequency-only baseline and examine sensitivity to forecast information. Finally, we inspect trip-time distributions to show how the aggregate gains are distributed across OD pairs.~~We organize the Mandl results in four parts. First, we report aggregate performance across variability levels and route quantities. Second, we inspect trip travel-time distributions to show how aggregate gains are distributed across OD pairs. Third, we compare full route redesign with a frequency-only baseline. Finally, we examine sensitivity to forecast information.

[4] Formal definitions are provided in App. D.

6.2.1. Aggregate Performance across Variability Levels

To select representative compromise solutions for the Mandl network under fluctuating travel-time, we compute the solutions on the Pareto front via the proposed Alg. 1. Then, we compute the Euclidean distance from each of those solutions to an ideal point representing the median observed values in the Pareto front, across the three objectives. The three solutions minimizing this distance are then chosen as the final candidates and are denoted as S1, S2, S3.

We use predicted travel times $\hat{t}_{ij}$ in Eq. (22) as input to the optimization problem, and we evaluate the resulting adaptive network against realized link travel times and prediction errors across multiple scenarios. Table 3 summarizes average performance across 30 independent realizations. It shows pronounced OD-pair travel-time improvements under high variability ($cv = 1.0$), while improvements remain significant even at lower variability levels. As expected, when variability is $cv = 0$, the central reference point (described in the previous paragraph) corresponds exactly to $\mathcal{G}_{\text{PT}}^{\text{static}}$. As travel-time variability increases, both 20% Improvement Share (IS_{20}) and Operational Cost Reduction (OCR) rise systematically, reflecting the increasing value of adaptive redesign under highly fluctuating conditions. At low variability ($cv = 0.05$), a sizable IS_{20} is still observed. This is because extreme link travel times are still possible in reality, even at low cv (the possibility of such extreme values is preserved by our choice of a Log-normal distribution §5.1. For a small-scale scenario like Mandl's network, such extremes inflate original trip times $t_k^0, k \in \mathcal{K}$. Accordingly, by exploring alternative trade-offs within the same objective space, the proposed approach can reduce passenger travel time while maintaining or improving operational costs, even in near-deterministic settings. Across all variability levels and redesign schemes, Route Overlap (RO) remains high, indicating that these improvements result from targeted adjustments rather than wholesale restructuring.

6.2.2. Trip Travel Time Distribution Analysis

zg~~To provide a conservative performance evaluation, we considered the "worst" realization, i.e., the seed that yielded the worst results in terms of the metrics of Table 3.~~For illustration, in this subsection we report the results for a single randomly chosen realization. Fig. 4 illustrates trip time distributions across varying coefficients of variation (cv). It highlights the benefits of our adaptive public transport planning method, as travel time uncertainty increases from minimal to high levels, against current network design approaches, represented by the original static network $\mathcal{G}_{\text{PT}}^{\text{static}}$ and Vermeir's network $\mathcal{G}_{\text{PT}}^{\text{ver}}$, in accordance to §6.1.1. At low variability levels (cv=0.05), less than 50% of OD pairs improve their travel time. This occurs because these static networks are already well-optimized for stable conditions, leaving little room for enhancement. Moreover, our optimization operates on predicted travel times that inevitably include errors. This is why, as illustrated in Fig. 1, we would not trigger any reconfiguration, faced with minor fluctuations in link travel times, such as those experienced with $cv = 0.05$.

However, as variability increases, the benefits of our approach ($\mathcal{G}_{\text{PT}}^{\text{adpt}}$) become pronounced. zg~~For instance, under high uncertainty (cv=1.0), our adaptive network $\mathcal{G}_{\text{PT}}^{\text{adpt}}$ improves by at least 30% the travel time of at least 40% of OD pairs compared to both $\mathcal{G}_{\text{PT}}^{\text{static}}$ and $\mathcal{G}_{\text{PT}}^{\text{ver}}$.~~For instance, under high uncertainty (cv=1.0), relative to the static design $\mathcal{G}_{\text{PT}}^{\text{static}}$, our adaptive network $\mathcal{G}_{\text{PT}}^{\text{adpt}}$ improves the travel time of about 30.6% of OD pairs by at least 20% (Table 3, demand-averaged IS_{20} over 30 realizations); relative to the travel-time-optimal design $\mathcal{G}_{\text{PT}}^{\text{ver}}$, the trip-time distribution in Fig. 4 shows that about 73.3% of travelers experience reduced travel time, with a reduction of at least 30% for roughly 30% of travelers (vs $\mathcal{G}_{\text{PT}}^{\text{ver}}$, single realization). zg~~This confirms the superiority of our proactive network redesign against current approaches, all keeping the network unchanged, in particular in highly fluctuating conditions.~~This suggests the potential benefit of our proactive network redesign over the selected static baselines, which keep the network unchanged, in particular in highly fluctuating conditions.

Beyond this single-realization view, a further OD-level improvement-rate analysis aggregated across all 30 seeds is provided in App. D.2.

6.2.3. Frequency-Only Baseline

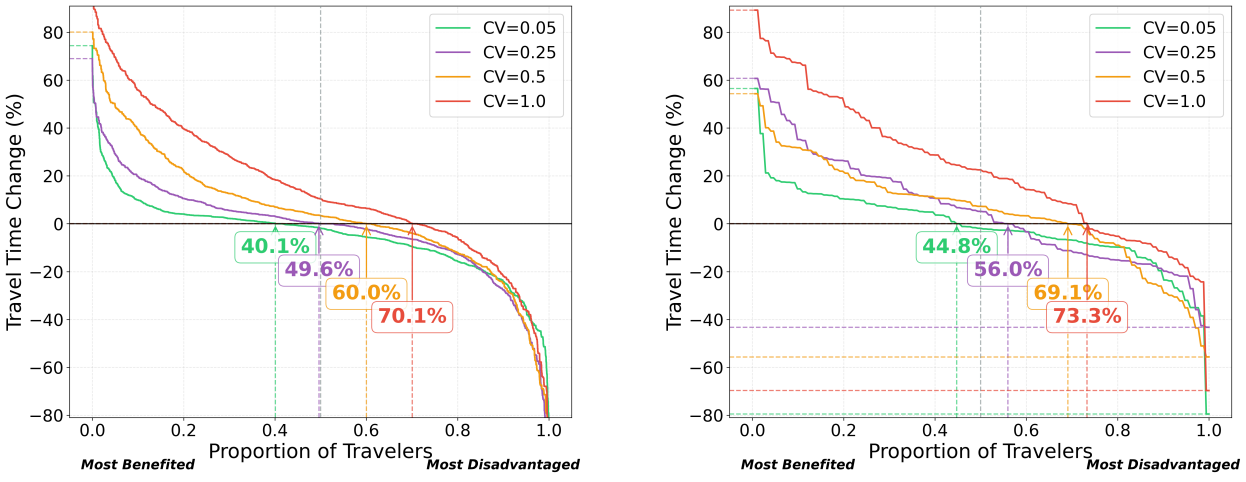
zgWe next examine whether the observed gains can be obtained by the fixed-route frequency-adjustment baseline $\mathcal{G}_{\text{PT}}^{\text{freq}}$ alone. Table 4 compares $\mathcal{G}_{\text{PT}}^{\text{freq}}$ with full network redesign, while Fig. 5 shows the corresponding Pareto fronts. In that figure, each point is one network design: the horizontal axis is the passenger travel-time cost (objective Z_1, normalized to $[0, 1]$) and the vertical axis is the operator cost (objective Z_2, total vehicle-hours), so designs closer to the lower-left corner are better. The frequency-only baseline $\mathcal{G}_{\text{PT}}^{\text{freq}}$ improves service under variability, but it captures substantially less of the OD-level improvement than full redesign; for example, at $cv = 1$, its 20%-improvement share is roughly half that of our method. The gap increases with travel-time variability, suggesting that the value of

Method	Metric	CV=0.05	CV=0.25	CV=0.50	CV=1.00
Frequency Only	20% improv. share (%)	3.96 ± 2.94	8.09 ± 4.35	12.94 ± 6.92	16.47 ± 9.53
	Op. Cost Red. (%)	12.85 ± 4.66	14.86 ± 5.78	19.27 ± 9.04	29.40 ± 15.04
	Route overlap (%)	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
Ours	20% improv. share (%)	8.25 ± 4.65	15.40 ± 5.66	24.63 ± 7.91	30.57 ± 11.60
	Op. Cost Red. (%)	13.28 ± 7.37	16.86 ± 6.74	24.06 ± 12.01	39.20 ± 12.94
	Route overlap (%)	83.79 ± 7.73	78.83 ± 6.65	77.53 ± 5.27	77.26 ± 5.19

FO: fixed-route frequency-only adjustment baseline $\mathcal{G}_{\text{PT}}^{\text{freq}}$; Ours: our full network redesign method. CV: coefficient of variation. 20% improvement share ($\text{IS}^{20\%}$-§6.1.3): fraction of OD pairs whose travel-time improvement is at least 20%; Op. Cost Red. (OCR): Operational Cost Reduction (reduction in total vehicle-hours; see §3.4.2). The 4-route static route-frequency solution is taken from Ahern et al. (2022); entries report mean ± standard deviation over 30 realizations.

Table 4
zgPerformance comparison of full network redesign and fixed-route frequency adjustment for the 4-route Mandl instance.

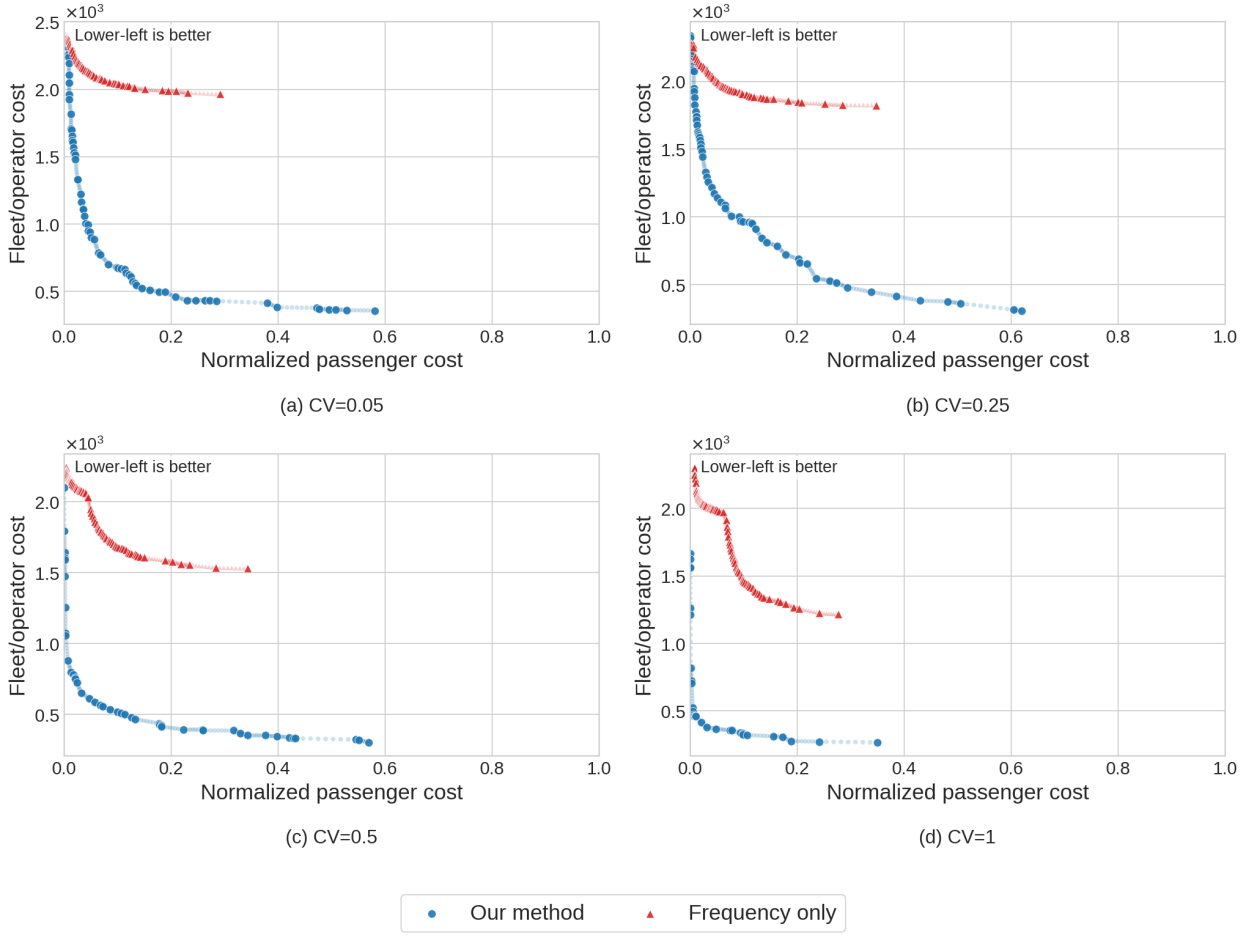
topology change is concentrated in precisely the settings where congestion patterns create larger spatial mismatches in the original network. While frequency-only adjustment keeps the route topology fixed, the proposed line redesign can reroute service away from deteriorated links.



(a) $\mathcal{G}_{\text{PT}}^{\text{adpt}}$ vs $\mathcal{G}_{\text{PT}}^{\text{static}}$

(b) $\mathcal{G}_{\text{PT}}^{\text{adpt}}$ vs $\mathcal{G}_{\text{PT}}^{\text{ver}}$ (Vermeir et al., 2021)

Figure 4: Trip Time Distribution Comparison Under Various Variability Conditions in a randomly chosen realization.



(a) CV=0.05

(b) CV=0.25

(c) CV=0.5

(d) CV=1

Figure 5: zgTravel-time, cost (Z_1, Z_2) Pareto fronts for the 4-route Mandl instance under four travel-time variability levels; lower-left is better.

6.2.4. Forecast-error Sensitivity

zgTo probe how much the redesign relies on forecast quality rather than on the travel-time variability it exploits, we hold each realized scenario $\{t_{ij}^{\omega}\}$ fixed and vary only the link times handed to the optimizer through the error-scale parameter α in Eq. (19). Here, $\alpha = 0$ gives perfect foresight, $\alpha = 1$ reproduces the empirical DCRNN error, and $\alpha > 1$ is an amplified-error stress test. At the Pareto-front level, we quantify this effect using two hypervolume-based summaries: forecast-value headroom, which measures the maximum gain attributable to using any forecast, and forecast value retained, which measures how much of that gain remains under an error-α forecast. Both are computed against the same realized scenario and aggregated over the 30 seeds with 95% confidence intervals; formal definitions and estimators are given in App. D.1.1. The resulting hypervolume analysis shows that forecast value is mainly present when variability creates redesign opportunities, and that the empirical forecast ($\alpha = 1$) retains essentially all available value even under moderate error amplification. Full results and detailed analysis are provided in App. D.1.2.

6.3. Large-scale validation on the Beijing network

We now move the analysis to the real Beijing network. To identify a representative solution for analysis, among the three solutions obtained as in Section 6.2, we only select here $S1$, i.e., the closest to the ideal point. The original and adjusted transit route networks are shown in Fig. 6.

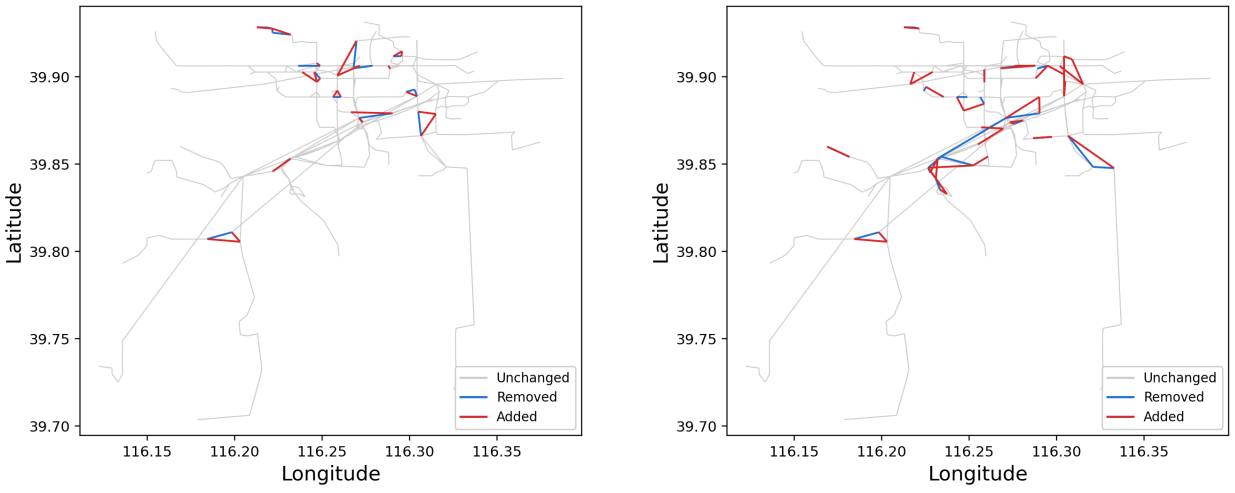
zgWe compare our method against a stop-skipping operational baseline on the Beijing network. For each variability level, links whose predicted travel time exceeds the original by more than a preset threshold are classified as degraded. The stop-skipping strategy then removes intermediate stops adjacent to degraded links whenever a direct shortcut is available, while leaving the overall line structure and vehicle allocation unchanged. The threshold thus controls aggressiveness: a 25% threshold skips stops whenever a link's predicted time grows by more than a quarter; a 100% threshold acts only when it doubles. This baseline represents a simple, local response to travel-time disruption, but structurally constrained: it cannot reroute vehicles or reassign fleet, and skipping stops may disconnect part of the demand from the network entirely. Table 5 reports route overlap (RO), operational cost reduction (OCR), travel-time improvement (TTI), and the resulting fractions of unreachable OD pairs and unserved demand for all threshold settings alongside our method.

Method	Thr. Skip threshold	CV Coeff. var.	RO Route overlap (%)	OCR Op. cost red. (%)	TTI Travel-time impr. (%)	Unreach. OD Unreachable OD (%)	Unserv. Dem. Unserved demand (%)
Stop-skip.	25%	0.00	91.16	1.24	3.26	4.89	3.89
		0.05	91.71	1.15	2.73	4.29	3.46
		0.25	56.02	6.58	12.43	29.12	27.39
		0.50	47.58	6.95	14.08	35.28	32.99
		1.00	42.18	12.71	14.71	38.93	36.04
	50%	0.00	95.52	0.61	−0.10	1.71	1.53
		0.05	95.24	0.63	1.94	1.71	1.53
		0.25	80.32	4.59	10.19	11.70	11.40
		0.50	56.13	8.69	19.42	28.63	27.56
		1.00	47.83	12.39	19.71	35.28	32.99
	100%	0.00	98.30	0.32	0.45	0.03	0.02
		0.05	98.30	0.35	0.49	0.03	0.02
		0.25	93.87	1.63	4.08	2.77	2.09
		0.50	78.57	6.68	17.77	12.23	11.61
		1.00	55.88	13.49	15.11	28.05	26.84
Ours	–	0.00	90.08	0.50	11.40	–	–
		0.05	87.34	1.53	11.60	–	–
		0.25	86.51	3.85	16.99	–	–
		0.50	86.39	12.02	18.44	–	–
		1.00	85.42	22.56	25.78	–	–

CV: coefficient of variation. Thr.: skip threshold used by the stop-skipping baseline. RO: Route Overlap. OCR: Operational Cost Reduction, measured as the reduction in total vehicle-hours; see §3.4.2. TTI: Travel-Time Improvement; see (21). Unreach. OD and Unserv. Dem. report the fraction of OD pairs and demand, respectively, whose connectivity is broken by stop-skipping. These metrics are not applicable to our method, which removes no stops and preserves full OD connectivity.

Table 5

zgComparison of our method and stop-skipping under varying travel-time variability on the Beijing network.



(a) $cv = 0$ (b) $cv = 1$

Figure 6: zg~~Original and adjusted transit route networks for the Beijing case study.~~Adapted Beijing transit route networks under low ($cv = 0$) and high ($cv = 1$) travel-time variability. Additional variability levels are shown in App. E.

In addition to the previous metrics, we also compute the following.

- **Avg. Travel Time Improvement (Avg.TTI):** percentage reduction in aggregate travel time across all OD pairs k relative to the original network:

$$\text{Avg.TTI} = \frac{\sum_{k\in\mathcal{K}}(t_k^0 - t_k^*)}{\sum_{k\in\mathcal{K}} t_k^0} \times 100\%, \text{ where } t_k^0, t_k^* \text{ are original and adaptive travel times.} \tag{21}$$

The results in Table 5 show that the original network exhibits some inefficiencies: the solution of our optimization yields an 11.40% Avg. TTI even at $cv = 0$. zg~~This reflects static re-optimisation of the incumbent network rather than prediction-driven adaptation. The incremental value of adaptation is the additional gain observed as cv increases.~~This gain at $cv = 0$ does not stem from prediction-driven adaptation: with no travel-time variability, there is nothing to forecast or to adapt to. Rather, it reflects a one-off re-optimisation of an incumbent network that, like most real-world bus systems, is not a mathematical optimum. Operational networks evolve incrementally over many years, with lines added or amended in response to historical demand and retained for service continuity, and are shaped by administrative, budgetary, and political constraints rather than periodic global re-optimisation, for which no optimization techniques are routinely used in practice (Durán-Micco and Vansteenwegen, 2022); as travel demand and land use shift, the fixed topology gradually drifts away from the efficient configuration. Our optimizer recovers part of this accumulated static inefficiency, which explains the 11.40% improvement even under nominal conditions. The genuine value of adaptation is therefore the additional gain observed as cv increases, beyond this one-time static correction. The benefit grows with variability: at higher cv, the algorithm produces designs increasingly different from the static original, achieving both travel-time improvements and line shortening, which in turn lowers operational cost.

zgTable 5 exposes a basic tension in stop-skipping: operational benefit comes at the cost of connectivity. Aggressive thresholds improve travel time on the reachable network but disconnect a large share of OD pairs as variability grows, whereas conservative thresholds keep connectivity yet act too rarely to help and still lose connectivity at high variability. No single threshold resolves this trade-off.

Our method achieves larger travel-time gains for two structural reasons. First, it redesigns entire line topologies rather than dropping individual stops, so vehicles can bypass congested corridors and serve demand more directly; this reaches reductions that stop-level changes cannot, since the bottleneck is usually the line's overall routing rather than a single link. Second, it jointly reallocates the fleet across lines, redeploying vehicles freed from shortened routes where they help most, while stop-skipping keeps the fleet fixed. Because no stops are removed, all OD pairs stay reachable and the reported TTI covers the full demand, making the gains comparable across variability levels without the connectivity caveat of every stop-skipping variant.

7. Conclusion

This paper introduced a novel framework for zg~~real-time PT proactive network redesign that maintains high performance under uncertain traffic conditions~~redesigning PT lines for the hour ahead from short-term forecasts, acting only when the forecast deviation from nominal conditions exceeds a preset tolerance, and keeping each redesign close to the network already in service. We formulated a multi-objective optimization problem balancing passenger travel time, operational costs, and network stability. Our predict-then-optimize approach relies on an NSGA-III-based optimization framework that proactively redesigns bus lines based on predicted line travel times.

We evaluated our approach against the inevitable prediction errors of well-established traffic-predicting models. zg~~To this aim, we fit the prediction error distribution via kernel density estimation.~~To this aim, we expose the optimizer to realistic forecast inaccuracy via empirical log-relative prediction error sampling drawn from a well-established deep learning-based prediction model (DCRNN). Experiments on Mandl's network and a real-world Beijing network demonstrate effective adaptation across varying levels of travel-time uncertainty. Results indicate that, faced with non-negligible link travel-time variability, the proposed approach zg~~yields substantial OD-level trip travel-time reductions, while achieving operational cost savings and preserving~~yields OD-level trip travel-time reductions, while achieving operational cost savings over the selected baselines and scenarios considered, and preserving a high degree of structural similarity with the baseline network. zg~~These findings highlight the importance of moving beyond static network design toward continuous, adaptive redesign in dynamic operating environments.~~Overall, the gains grow with travel-time variability and come with high route overlap, indicating that the benefit does not rely on large structural change. A

natural extension is to enrich the forecast-error model with explicit spatial and temporal correlation, allowing network-wide common shocks to be represented. zgA further direction is to relax the uncongested user-optimal assignment (Observation 1) by incorporating in-vehicle crowding, for instance through a capacitated or congestion-dependent (frequency-based common-line) passenger assignment; this would let the redesign account for the load it induces on each line and report crowding-aware rather than purely uncongested travel-time gains.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The Mandl benchmark instance is publicly available in the cited literature. The Beijing network data are derived from third-party sources cited in the paper. The problem instances and source code will be released in a public repository upon acceptance.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, ChatGPT (GPT-5.1, 2025 release; OpenAI, San Francisco, CA, USA) was used to refine the language of the manuscript and ensure compliance with formatting, grammar, and British English conventions. After using these tools, the authors reviewed and edited the manuscript as needed and take full responsibility for the final content of the publication.

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A. Proof and Discussion of Proposition 2

Proof. Setting $A = \mathcal{E}^{(k)}$, $B = \mathcal{E}^{(0)}$, and $C = \mathcal{E}^{(k+n)}$. The Jaccard distance between two sets X and Y is

$$d_J(X,Y) = 1 - \frac{|X \cap Y|}{|X \cup Y|} = 1 - J(X,Y).$$

Levandowsky and Winter (Levandowsky and Winter, 1971) noted that this dissimilarity measure satisfies the triangle inequality on finite sets, so that d_J may be regarded as a distance function. Hence,

$$d_J(A,C) \leq d_J(A,B) + d_J(B,C).$$

Substituting $d_J(X,Y) = 1 - J(X,Y)$ into the above inequality yields

$$1 - J(A,C) \leq [\,1 - J(A,B)\,] + [\,1 - J(B,C)\,],$$

which is equivalent to

$$J(A,C) \geq J(A,B) + J(B,C) - 1.$$

zg, ~~gives the desired result.~~which gives the desired result. ■

Remark. This implies that if two consecutive designs exhibit a similarity of $\geq 90\%$ relative to the original (consistent with our numerical results), the similarity between these two consecutive designs is guaranteed to be at least 80%.

B. Stochastic Modelling of Travel Times and Prediction Errors

B.1. Parameter Derivation for Log-Normal Distribution

Proof of Proposition 3. For a lognormal random variable $X \sim \text{LogNormal}(\mu, \sigma^2)$, the median is $\text{Med}(X) = \exp(\mu)$, so specifying the median m yields

$$\mu = \ln(m).$$

The mean and variance of X are

$$\text{E}[X] = \exp\left(\mu + \tfrac{1}{2}\sigma^2\right), \qquad \text{Var}(X) = \left(\exp(\sigma^2) - 1\right)\exp\left(2\mu + \sigma^2\right).$$

Hence the coefficient of variation is

$$\begin{aligned} cv &= \frac{\sqrt{\text{Var}(X)}}{\text{E}[X]} \\ &= \frac{\sqrt{\left(\exp(\sigma^2) - 1\right)\exp(2\mu + \sigma^2)}}{\exp(\mu + \tfrac{1}{2}\sigma^2)} \\ &= \sqrt{\exp(\sigma^2) - 1}. \end{aligned}$$

Equating this expression with the prescribed value of cv gives

$$cv^2 = \exp(\sigma^2) - 1 \quad \Longrightarrow \quad \sigma^2 = \ln\left(1 + cv^2\right) \quad \Longrightarrow \quad \sigma = \sqrt{\ln\left(1 + cv^2\right)}.$$

■

Table 6
One-hour-ahead prediction error on benchmark datasets.

Dataset	Model	MAE	RMSE
METR-LA	DCRNN	3.590	7.607
	MTGNN	3.490	7.230
Q-Traffic	DCRNN	3.932	6.030
	MTGNN	2.973	4.742

B.2. zgTraffic Speed Prediction and Error Estimation

To generate realistic stochastic prediction errors, we employ the Diffusion Convolutional Recurrent Neural Network (DCRNN) (Li et al., 2017) to model spatiotemporal traffic dependencies. Two benchmark datasets are utilized: **METR-LA** (4 months of highway data from 207 sensors) as a proxy for Mandl's network, and **Q-Traffic** (61 days of Beijing 6th Ring Road segments) for the real-world case study. We split both datasets, chronologically, as recommended by Li et al. (2017) and Lu, Lv, Xie, Du, Xiong, Sun and Wang (2022), respectively: We train each prediction model on the first part of the chronologically ordered data and measure prediction errors on the last part, i.e., the test set. Table 6 reports the one-hour-ahead prediction errors used in our experiments.

The prediction models output traffic speeds. Since the downstream public-transport redesign model operates on travel times, the extracted prediction residuals must be expressed in travel-time space. If a link length d is available, the realized and predicted travel times are $T = d/v$ and $\hat{T} = d/\hat{v}$, respectively. The corresponding log-relative travel-time residual is therefore $\eta = \log(\hat{T}) - \log(T)$. For a fixed link length, this is equivalently written as $\eta = \log(v) - \log(\hat{v})$, since the length term cancels out algebraically. This equivalence is particularly useful for **METR-LA**, where reliable segment lengths are not directly available for all sensor pairs. In this case, we compute the travel-time log-relative residual directly from the observed and predicted speeds. For **Q-Traffic**, where segment lengths are explicitly provided, the same residual can be obtained either after converting speeds into travel times or directly through the equivalent speed-based expression. The resulting residual pool is then used to generate noisy travel-time forecasts as described in App. B.3.

B.3. zgEmpirical Log-Relative Residual Sampling for Travel-Time Prediction

We pool all empirical prediction residuals on the test sets, across links and time steps, to construct a representative empirical error pool. Since the downstream experiments transfer prediction errors across links and networks with different travel-time scales, we sample log-relative residuals rather than absolute residuals. Specifically, for each realized travel time T and its prediction $\hat{T}$, we define $\eta = \log(\hat{T}) - \log(T) = \log\left(\hat{T}/T\right)$. This log-ratio formulation is appropriate for heteroscedastic prediction errors whose magnitude scales with the level of the predicted variable (Tofallis, 2015); in our setting, this is consistent with traffic scenarios in which the coefficient of variation changes the scale of realized travel times. Finally, independent samples η_{ij}^{ω} are drawn with replacement from this empirical pool for each link (i, j) and scenario ω. Given the realized travel time t_{ij}^{ω} generated under the prescribed traffic-variability level, the corresponding additive travel-time error is defined as $\epsilon_{ij}^{\omega} = t_{ij}^{\omega}\left(\exp!\left(\alpha\eta_{ij}^{\omega}\right) - 1\right)$. The predicted link travel times $\hat{t}_{ij}^{\omega}$, used to make decisions via solving (5)-(14), are then computed by adding ϵ_{ij}^{ω} to the actual travel time t_{ij}^{ω}, according to (22). We do this in both Mandl and Beijing scenarios.

Moreover, because errors are sampled independently for each link and scenario, this first-order model abstracts away the spatial and temporal correlation of real forecast errors, such as corridor-level or incident-level common shocks. The reported results should therefore be read as an evaluation under sampled independent empirical residual scenarios, and correlation-aware forecast-error modeling remains an important extension.

$$\hat{t}_{ij}^{\omega} = t_{ij}^{\omega} + \epsilon_{ij}^{\omega},\ \epsilon_{ij}^{\omega} = t_{ij}^{\omega}\left(\exp!\left(\alpha\eta_{ij}^{\omega}\right) - 1\right) \tag{22}$$

Diagnostic check on traffic variability. For the Mandl experiments, we further report the realized-to-nominal travel-time ratio $R_{ij}^{\omega} = t_{ij}^{\omega}/t_{ij}^{0}$, where t_{ij}^{0} is the nominal link travel time and t_{ij}^{ω} is the realized travel time under scenario ω. Table 7 summarizes the distribution of R under different traffic-variability levels. The results show that increasing cv mainly amplifies the dispersion and upper tail of realized travel times. Small-cv scenarios remain close to nominal

Table 7
Distribution of realized-to-nominal travel-time ratios in the Mandl experiments.

cv	Mean(R)	Std(R)	P95(R)
0.00	1.000	0.000	1.000
0.05	0.988	0.047	1.076
0.25	0.961	0.228	1.436
0.50	0.973	0.456	2.001
1.00	1.090	0.949	3.396

conditions, whereas large-cv scenarios generate severe congested realizations, as reflected by the increasing standard deviation and 95th percentile of R. In these diagnostics, the prediction-error scale is fixed at $\alpha = 1.0$, corresponding to the empirical error magnitude extracted from the trained prediction model.

C. Algorithmic Framework and Evaluation

C.1. Parameter values for the proposed Algorithm 1

When computing adaptive network $\mathcal{G}_{\text{PT}}$ using the solution method described in § 4, we set the algorithmic parameters introduced therein as follows. The minimum path-overlap threshold is set to $J_{\min} = 0.5$ (§4.2.1), the crossover and mutation probabilities to $P_c = 0.85$ (§4.2.2) and $P_m = 0.35$ (§4.2.3), and the number of reference points to $N_{\text{ref}} = 5^3 = 125$ (§4.2.4), respectively.

C.2. Performance Metrics for Multi-Objective Optimization

We evaluate algorithm performance using three standard metrics for multi-objective optimization: Hypervolume (HV), Inverted Generational Distance (IGD), and Generational Distance (GD). The HV metric quantifies the volume of the objective space dominated by the Pareto solutions:

$$\text{HV}(X, P) = \Lambda\left(\bigcup_{x \in X} v(x, P)\right) \tag{23}$$

where X is the approximation set, P is the reference point, and $v(x, P)$ is the hypervolume dominated by solution x relative to P. Higher HV values indicate better coverage of the objective space. To ensure a fair comparison across algorithms, we report the HV metric calculated relative to a unified reference point. This reference point is constructed by collecting all solutions generated by all algorithms for each objective and selecting the worst (largest) value observed for each objective. The IGD metric measures the closeness of the approximation set to the true Pareto front:

$$\text{IGD}(X, \mathcal{P}^*) = \frac{\sum_{x^* \in \mathcal{P}^*} d(x^*, X)}{|\mathcal{P}^*|} \tag{24}$$

where $\mathcal{P}^*$ is a set of reference points on the true Pareto front, $d(x^*, X)$ is the minimum distance from x^* to any point in X, and $|\mathcal{P}^*|$ is the cardinality of $\mathcal{P}^*$. Lower IGD values indicate better convergence to the true Pareto front. The GD metric complements IGD by measuring how far the obtained solutions are from the true Pareto front:

$$\text{GD}(X, \mathcal{P}^*) = \frac{\sum_{x \in X} d(x, \mathcal{P}^*)}{|X|} \tag{25}$$

where $d(x, \mathcal{P}^*)$ is the minimum distance from a solution x to the true Pareto front $\mathcal{P}^*$, and $|X|$ is the cardinality of the approximation set. Lower GD values indicate better convergence of the approximation set to the true Pareto front. zgAs the true Pareto front $\mathcal{P}^*$ is unknown for the instances considered, we approximate it, following standard practice in multi-objective optimization, by the non-dominated subset of the union of all solutions produced by all compared algorithms across all runs at a given variability level. The HV reference point is constructed analogously from the worst objective values observed across all algorithms (as described above), so that HV, IGD, and GD are all evaluated against a common, algorithm-agnostic reference.

C.3. Implementation Details of the NSGA-III Environmental Selection

The algorithm adopts the reference-point-based environmental selection mechanism of NSGA-III (Deb and Jain, 2014) to maintain diversity across the objectives (5). Since all objectives are formulated as minimization problems, normalization is performed on the combined population $R_t = P_t \cup Q_t$. For each objective $j \in \{1, 2, 3\}$, objective-wise lower and upper bounds are computed as $z_j^{\min} = \min_{x \in R_t} Z_j(x)$ and $z_j^{\max} = \max_{x \in R_t} Z_j(x)$, and each objective is linearly scaled to $[0, 1]$ accordingly.

A set of uniformly distributed reference points on the unit simplex is generated following the simplex-lattice design in Deb and Jain (2014). As in standard multiobjective optimization, we say that solution x dominates y if x is no worse than y in all objectives and strictly better in at least one. We employ non-dominated sorting, which partitions the **combined population** $R_t = P_t \cup Q_t$ (comprising both parent and offspring individuals) into a sequence of Pareto fronts $F = \{F_1, F_2, \ldots, F_l\}$. Here, F_1 contains the globally non-dominated solutions F_2 contains solutions dominated exclusively by members of F_1, and F_l represents the *critical front* where the cumulative count of solutions first meets or exceeds the population capacity *PopSize*. Each solution is associated with its nearest reference point by computing perpendicular distances from the solution to the reference lines in the normalized space. The environmental selection process (line 10 of Alg. 1) adds complete Pareto fronts F_1 through F_{l-1} sequentially; for the last front F_l, slots are filled by prioritizing solutions associated with reference points that have fewer assigned solutions to ensure diversity.

Algorithm 2 Feasibility-Oriented Path Completion for Line Repair

Input: Substrate graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, endpoints (u, v), visited node set $\mathcal{V}^{\ell}_{\text{used}}$
Output: Connecting path $\mathcal{P}_{u \to v}$ or $\emptyset$

$visited \leftarrow \{u\} \cup \mathcal{V}^{\ell}_{\text{used}}$, $queue \leftarrow \langle u \rangle$, $parent(\cdot) \leftarrow \emptyset$
while *queue* is not empty **do**
  $i \leftarrow queue$.pop_front()
  if $i = v$ **then**
    break
  end if
  for each neighbor j of i in $\mathcal{G}$ **do**
    if $j \notin visited$ **then**
      $visited \leftarrow visited \cup \{j\}$
      $parent(j) \leftarrow i$
      $queue$.push_back(j)
    end if
  end for
end while
if $v \notin visited$ **then**
  return $\emptyset$
end if
$\mathcal{P}_{u \to v} \leftarrow$ Backtrack from v to u using $parent(\cdot)$
return $\mathcal{P}_{u \to v}$

C.4. Algorithm Performance Evaluation

Algorithmic performance is evaluated via the Hypervolume (HV), the Inverted Generational Distance (IGD), and the Generational Distance (GD), which are standard performance metrics in multi-objective evolutionary optimization as surveyed in (Riquelme, Von Lücken and Baran, 2015).[5] The Hypervolume (HV) evaluates both convergence and diversity by measuring the dominated volume in the objective space, where higher values indicate better performance (↑). In contrast, GD and IGD are distance-based indicators that assess convergence to a reference Pareto front, with lower values indicating better convergence (↓).

To evaluate the effectiveness of NSGA-III for zg~~real-time~~forecast-triggered PT redesign, we compare it with NSGA-II (Deb, Pratap, Agarwal and Meyarivan, 2002) and hybrid multi-objective algorithms that combine non-dominated sorting with variable neighborhood descent, based on Mandl's Swiss network. Specifically, we consider two Non-dominated Sorting Variable Neighborhood Descent (NSVND) variants inspired by the integration of NSGA-II with variable neighborhood search strategies (Hansen and Mladenović, 2001). The first variant (NSVND-II) follows the

[5] Formal definitions and computational details are provided in App. C.2.

Performance Metric	NSGA-II	NSGA-III	NSVND-II	NSVND-III
Inverted Generational Distance (IGD, lower is better)	347.64	**22.63**	38.90	34.80
IGD Change[a]	–	**-325.01**	-308.74	-312.84
Generational Distance (GD, lower is better)	75.36	**17.17**	24.40	22.80
GD Change[a]	–	**-58.19**	-50.96	-52.56
Hypervolume (HV, higher is better) [b]	30.52×10^9	31.49×10^9	31.23×10^9	31.31×10^9
HV Change[a]	–	**0.97**$\times 10^9$	0.71$\times 10^9$	0.79$\times 10^9$

[a] Improvement/Change are computed relative to NSGA-II (baseline).
[b] HV uses a single fixed reference point shared by all algorithms.

Table 8
Multi-objective algorithm comparison under noise-free travel times ($cv = 0$).

NSGA-II framework while replacing crossover and mutation with variable neighborhood descent, whereas the second variant (NSVND-III) further incorporates an adaptive neighborhood selection mechanism to enhance exploration of the Pareto frontier.

Table 8 presents an algorithm performance comparison for adaptive network optimization assuming no travel time deviation from the nominal ones (cv=0). Although no adaptive redesign is required when $cv = 0$, this setting provides a noise–free benchmark for comparing optimization algorithms. NSGA-III maintains an optimal balance between solution diversity (high hypervolume) and convergence precision.

D. Network Performance Metrics

This appendix details the performance indicators used in § 6.1.3.

For each realization $\omega \in \Omega$ of stochastic travel times, we define the following performance metrics: **20% Improvement Share (IS_{20})**, **Operational Cost Reduction (OCR)**, and **Route Overlap (RO)**. Since these metrics vary with the realized scenario ω, they are treated as random variables. In our numerical results, we report their empirical averages (and standard deviations) over 30 independent realizations.

- **20% improv. share (%):** The proportion of OD pairs experiencing a travel time reduction of at least 20% under realization ω.

$$\mathrm{IS}_{20}(\omega) = \frac{\sum_{k\in\mathcal{K}} \mathbb{I}\left(\frac{t_k^0 - t_k^*(\omega)}{t_k^0} \geq 20\%\right)}{|\mathcal{K}|} \times 100\%, \text{ where } \mathbb{I}(\cdot) \text{ is the indicator function.} \tag{26}$$

  Here t_k^0 and $t_k^*(\omega)$ denote the original and adaptive travel times for OD pair k, respectively. Since both depend on random travel-time realizations, IS_{20} is a random variable.

- **Operational Cost Reduction (OCR)**

$$\mathrm{OCR}(\omega) = \frac{\sum_{\ell\in\mathcal{L}} n^\ell t^\ell - \sum_{\ell\in\mathcal{L}} n^{\ell,*}(\omega) t^{\ell,*}(\omega)}{\sum_{\ell\in\mathcal{L}} n^\ell t^\ell} \times 100\%. \tag{27}$$

  Here t^ℓ and $t^{\ell,*}(\omega)$ are the realized travel times of route ℓ in the original and adaptive networks, respectively, and n^ℓ and $n^{\ell,*}$ are the corresponding vehicle allocations. Because t^ℓ and $t^{\ell,*}(\omega)$ vary with the realization, OCR is also a random variable.

- **Route Overlap (RO)** To quantify structural similarity between the original and adaptive networks, we use the Jaccard index:

$$\mathrm{RO}(\omega) = \frac{\left|\mathcal{E}_{\mathrm{PT}} \cap \mathcal{E}_{\mathrm{PT}}^*(\omega)\right|}{\left|\mathcal{E}_{\mathrm{PT}} \cup \mathcal{E}_{\mathrm{PT}}^*(\omega)\right|} \times 100\%. \tag{28}$$

Here $\mathcal{E}_{\mathrm{PT}}$ and $\mathcal{E}^*_{\mathrm{PT}}(\omega)$ denote the realized edge sets of the original and adaptive networks, respectively. Since these sets depend on the travel-time realization, RO is likewise a random variable.

The numerical results therefore report the empirical means ± standard deviations of IS_{20}, OCR, and RO over the 30 realizations $\omega \in \Omega$, providing a stochastic assessment of adaptive-network performance.

D.1. zgForecast-error Sensitivity Analysis

zgThis appendix provides the formal basis and the full results for the forecast-error sensitivity analysis summarized in the main text. App. D.1.1 defines the two hypervolume-based summaries used to quantify forecast value (the forecast-value headroom and the forecast value retained). App. D.1.2 then applies these metrics to the Mandl instance and reports the detailed results across variability levels and error scales α.

D.1.1. zgForecast-value metrics

zgWe quantify whether the forecast helps the redesign, and by how much. Holding the realized scenario $\{t^{\omega}_{ij}\}$ fixed, we vary only the forecast link times $\hat{t}^{\omega}_{ij}(\alpha)$ handed to the optimizer (Eq. (19)) and score the returned Pareto front on the true realized times t^{ω}_{ij}; since only the error scale α changes, any difference is attributable to forecast quality alone. Each front is graded by its hypervolume HV, the volume it dominates in the space of the three performance metrics (IS_{20}, OCR, RO) defined in §6.1.3, measured from the single fixed reference point of App. C.2; a larger HV is better and needs no weighting of the three metrics. For each seed $s \in \{1, \dots, S\}$ (with $S = 30$ random seeds) and each variability level c we compute three fronts:

- No forecast: nominal link times t^0_{ij}, giving the floor $HV^{\mathrm{NF}}_{s,c}$ (lower bound).
- Perfect foresight ($\alpha = 0$): the true realized times $t^{\omega}_{ij} = \hat{t}^{\omega}_{ij}(0)$, giving the ceiling $HV^{\mathrm{PF}}_{s,c} \equiv HV_{s,c,0}$ (upper bound).
- Error-α forecast: the forecast $\hat{t}^{\omega}_{ij}(\alpha)$ of Eq. (19) ($\alpha = 1$ the empirical DCRNN level, $\alpha > 1$ a stress test), giving the intermediate hypervolume $HV_{s,c,\alpha}$.

Because a seed fixes both the realized scenario and the sampled errors η, each $HV_{s,c,\alpha}$ is a single realization, and all reported 95% confidence intervals reflect variation across the S seeds. We measure the climb from floor to ceiling in two steps.

Step 1: is there anything to gain? The forecast-value headroom $H_{s,c} = \left(HV^{\mathrm{PF}}_{s,c} - HV^{\mathrm{NF}}_{s,c}\right)/HV^{\mathrm{PF}}_{s,c}$ is the floor-to-ceiling gap as a fraction of the ceiling: $H \approx 0$ means even a perfect forecast barely beats nominal times, whereas a large H signals real value to capture. We report the per-seed mean $\bar{H}_c$ with a 95% confidence interval.

Step 2: how much do we actually capture? The forecast value retained $R_{s,c,\alpha} = \left(HV_{s,c,\alpha} - HV^{\mathrm{NF}}_{s,c}\right)/\left(HV^{\mathrm{PF}}_{s,c} - HV^{\mathrm{NF}}_{s,c}\right)$ places an error-α forecast on that same scale: $R = 1$ matches perfect foresight, $R = 0$ matches no forecast, and $R < 0$ is worse than no forecast. Since this gap can be tiny for a single seed (making per-seed ratios erratic, unlike $H_{s,c}$, whose denominator $HV^{\mathrm{PF}}_{s,c}$ stays bounded away from zero), we aggregate it as a ratio of averages,

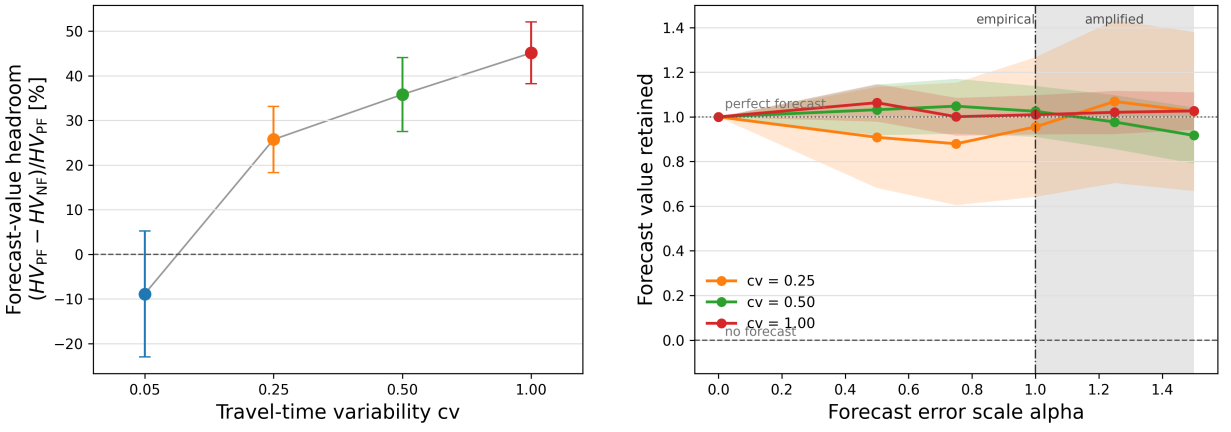
$$\bar{R}_{c,\alpha} = \frac{\frac{1}{S}\sum_s \left(HV_{s,c,\alpha} - HV^{\mathrm{NF}}_{s,c}\right)}{\frac{1}{S}\sum_s \left(HV^{\mathrm{PF}}_{s,c} - HV^{\mathrm{NF}}_{s,c}\right)}, \tag{29}$$

with a delta-method 95% confidence interval and $\bar{R}_{c,0} = 1$ by construction. At the lowest variability ($cv = 0.05$) the floor and ceiling are statistically indistinguishable (Fig. 7a), so R is undefined and we omit this level from Fig. 7b.

D.1.2. zgForecast-value results

zgFig. 7 reports the forecast-value metrics of App. D.1.1 for the 4-route Mandl instance. By construction, these metrics separate two otherwise-conflated effects: the redesign opportunity created by travel-time variability (the headroom), and the accuracy with which the forecast exploits it (the retained value). Fig. 7(a) shows that the headroom is negligible at $cv = 0.05$ and grows with variability, reaching roughly half of the perfect-foresight hypervolume at $cv = 1.0$: a forecast is valuable primarily when variability creates a genuine redesign opportunity, whereas in stable conditions even an oracle adds little because the nominal times t^0_{ij} already describe the operating state. Fig. 7(b) shows

that the empirical forecast ($\alpha = 1$) retains essentially all of this available value, and moderate amplification ($\alpha = 1.5$) causes no visible collapse, consistent with the intended robustness claim. **The method's gains are therefore driven mainly by the redesign opportunities that variability creates, not by a dependence on high-precision forecasts.**



(a) zgForecast-value headroom $\bar{H}_c$ vs. cv (mean over 30 seeds; 95% CI bars). Headroom is the share of the perfect-foresight hypervolume attributable to using any forecast.

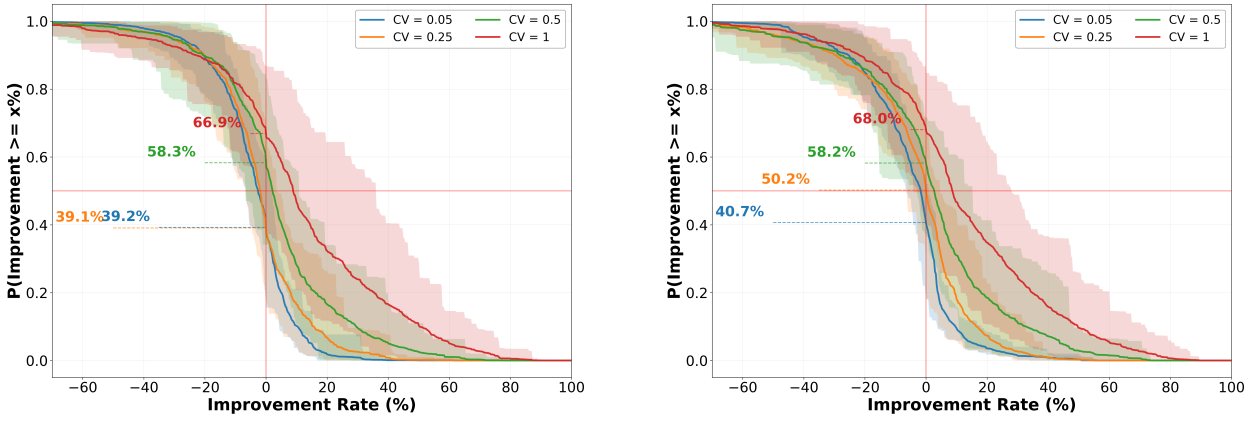
(b) zgForecast value retained $\bar{R}_{c,\alpha}$ vs. error scale α (ratio-of-means; shaded 95% CI). $\alpha = 1$ is the empirical DCRNN error; $\alpha > 1$ is a stress test.

Figure 7: zgHypervolume-based forecast value on the Mandl redesign (mean over 30 seeds): (a) headroom available from any forecast as variability grows; (b) fraction of headroom retained as forecast error is amplified. The method retains essentially all available value at the empirical error level and degrades only under severe amplification.

D.2. zgExceedance Probability of OD-Level Improvement Rates by Bus-Line Quantity

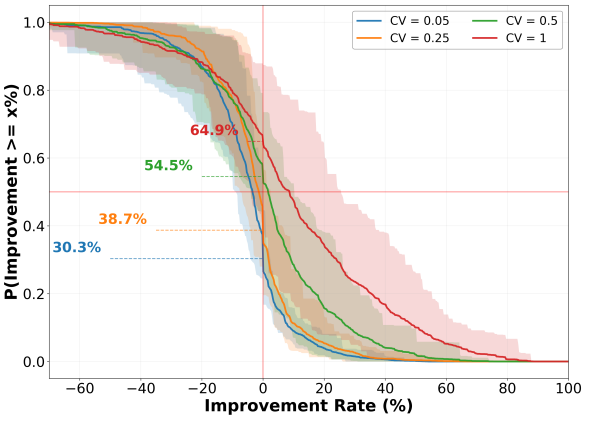
aaFor each variability level $cv = 0.05, 0.25, 0.5, 1$ and each random seed $\omega \in \Omega$, we compute the empirical complementary cumulative distribution function (CCDF) of the percentage travel-time improvement over the set of OD pairs. Thus, for a threshold x, the CCDF gives the fraction of OD pairs whose percentage improvement is larger than x. The solid lines in Fig. 8 report the pointwise average of these empirical CCDFs over the 30 seeds. The figure shows that proactive redesign becomes beneficial only under large travel-time variability, namely for ($cv \geq 0.5$). For lower variability levels, it worsens travel times for a larger fraction of OD pairs than it improves.

aaFig. 8 reports the trimmed-mean exceedance curves of OD-level improvement rates. For each seed ω, the OD-level improvement rates are first sorted to form a per-seed quantile curve $Q_\omega(p)$, where p denotes the exceedance probability. The extreme seed curves are then removed and the remaining per-seed quantile curves are averaged pointwise. Thus, the plotted line is a trimmed mean of per-seed quantile curves, rather than a curve computed from pooled OD pairs across all seeds. Across route quantities, the curves make it possible to compare both the intensity of improvements and the share of OD pairs receiving a given minimum percentage reduction under different variability levels.

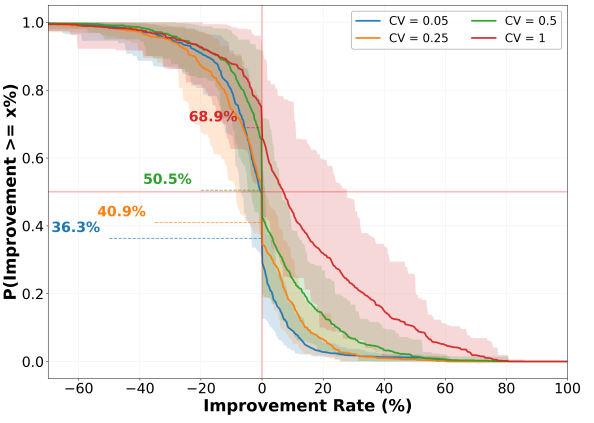


(a) 4 routes

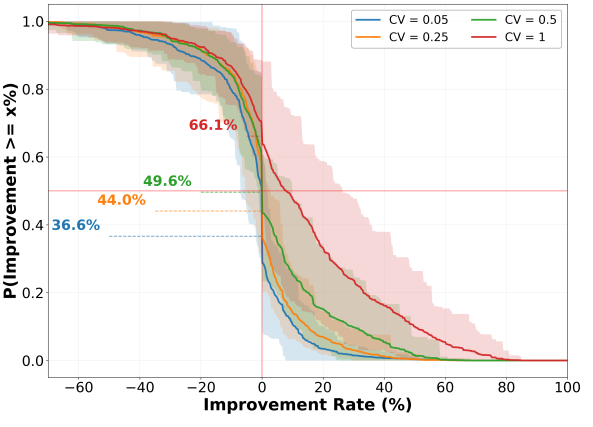
(b) 6 routes



(c) 8 routes



(d) 10 routes

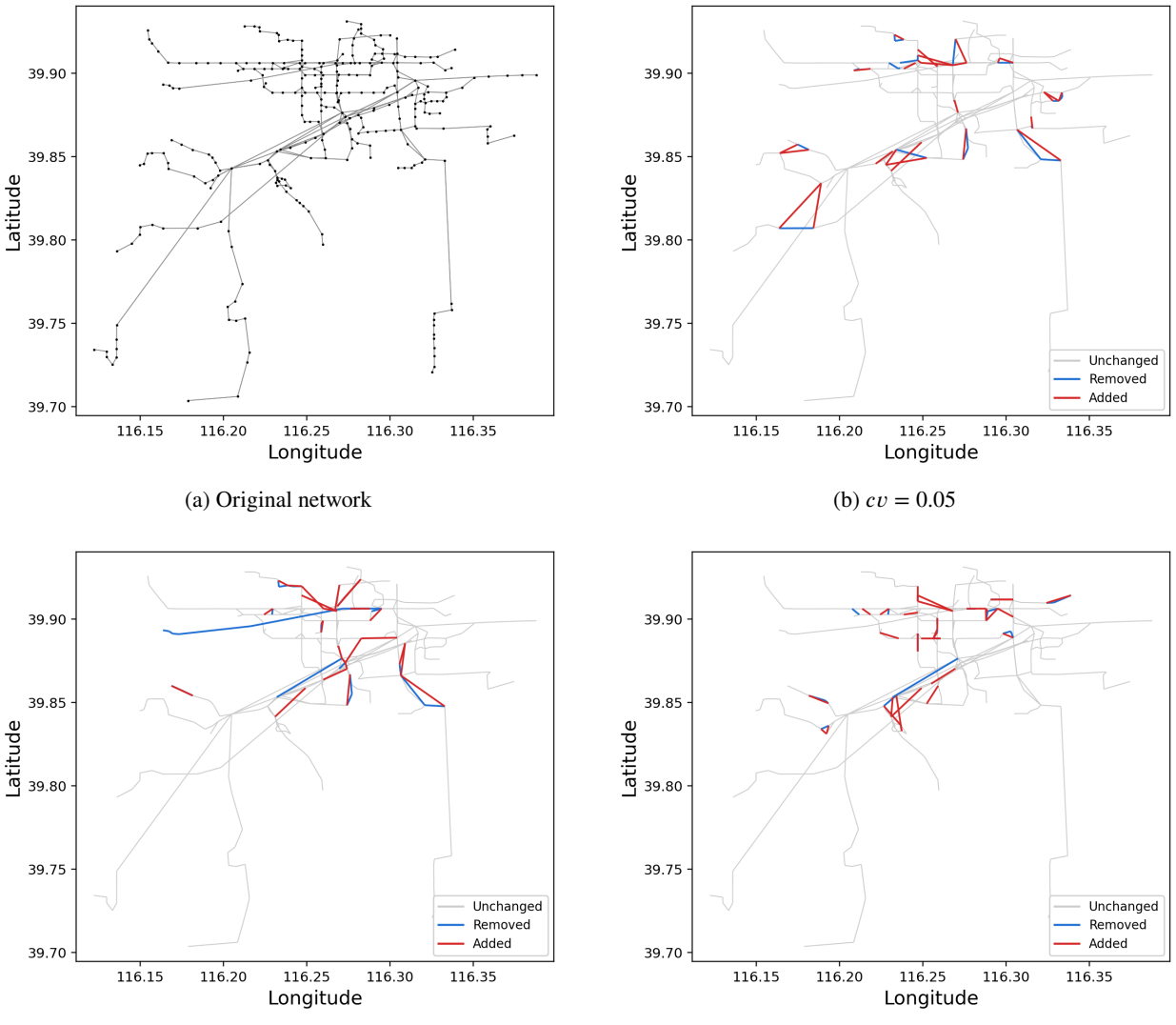


(e) 12 routes

Figure 8: zgTrimmed-mean exceedance probability curves of OD-level improvement rates under different route quantities and variability levels. Each line is computed by first forming one per-seed quantile curve $Q_\omega(p)$ and then averaging the remaining curves after trimming extreme seed curves.

E. Additional Beijing Network Visualisations

zgFig. 9 complements the Beijing case-study visualisation in Fig. 6 by reporting the original network and the adapted route layouts at intermediate variability levels. These additional panels show that the redesign remains spatially targeted as cv increases: routes are adjusted around affected corridors while preserving the broad structure of the original service network.



(a) Original network

(b) $cv = 0.05$

(c) $cv = 0.25$

(d) $cv = 0.5$

Figure 9: zgBeijing transit route networks: original design and adapted networks under $cv \in \{0.05, 0.25, 0.5\}$. The $cv = 0$ and $cv = 1$ cases are shown in the main text (Fig. 6).